OpenSourceRail

Open Engineering Platform

An illustrated guide to the railway, its people and its open engineering work



Wheel-supported reference and candidate-bound research. Manufacturing, physical and operating acceptance remain open.

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I. An open railway for people

Purpose, passenger experience and the design reference.

1.1 A note to the reader

This book is for people who want to understand the railway as a whole: the passenger experience, the city it serves, the train and guideway, the workshops that could make and maintain it, and the decisions still required before anyone could build or operate it. It is also an account of the open engineering platform that connects those subjects.

OpenSourceRail begins with a practical question. Can a comfortable, accessible urban railway be designed so that cities can understand its complete cost and develop lasting local capability to make, repair and improve it? The project explores a relatively simple wheel-supported train and station-powered propulsion, while allowing other arrangements to compete when they serve passengers better. It treats design, construction, operation and renewal as one problem. A cheaper train has little value if it needs an unaffordable station, an unrepairable part or an unworkable service pattern.

The present reference is a six-car train of 17-metre cars, with passive steel wheels on conventional rails carried by concrete. Each car is planned for 44 fixed seats and two wheelchair places, with no planned standing. These are development choices, not promises of an approved train. Propulsion, braking, headway and the final route must be established for each complete candidate. Safety, passenger protection and accessibility are requirements throughout.

What the reader will find

Part I explains the purpose and the passenger journey. Part II follows a city proposal into a worked Samawah corridor revision. Part III describes the vehicle, doors, power, braking, stations and structures. Part IV asks how the railway could be made, costed, controlled and maintained. Part V explains the engineering tools, the evidence already gained, the gaps that remain and the route through v0.7. Short definitions and suggested further reading close the book. The PDF outline provides chapter navigation.

Illustrations are labelled as concepts or engineering views. A concept image can help explain a proposal without proving dimensions, performance or qualification. Calculated results are presented with their assumptions and limits. The published v0.6.1 release is a software release: it makes planning, research and review tools available, while physical manufacture and passenger operation remain subject to separate engineering and approval work.

Reading the evidence

Four descriptions recur in this book. A proposal is a design intention. A calculated result follows from a stated model and its inputs. A verified software result shows that a workflow ran and its outputs can be checked. A physically accepted result would require appropriate measurements, approved limits and independent review. The first three can help design a railway; none automatically becomes the fourth.

The current railway is not a single finished object. There is a selected development reference and there are research alternatives. When a vehicle, station or propulsion arrangement changes, its new geometry, mass, forces, power, heat, construction and cost consequences must travel together. The book calls this a candidate. Figures from a different candidate are useful for comparison, but cannot be inserted into the selected train as though they had been measured there.

The machine-readable source receipt beside the PDF records the exact inputs, illustrations and book file. It is for verification, so long file identities and execution logs are kept out of these chapters. Readers who want the detailed methods can follow the named project guides in the final chapter.

Edition: 13 November 2026. The v0.6.1 Open Engineering Platform software release is published; v0.7 engineering and integration continue.

1.2 Why build an open railway

Urban transport shapes where people can live, study and work. A new railway can shorten journeys and make a city more accessible, but it also commits the city to decades of structures, vehicles, energy use, staffing, renewal and repairs. OpenSourceRail starts from that lifetime commitment. Its aim is an affordable railway that carries useful numbers of passengers in ordinary service and leaves the owner able to understand and improve what it has built.

The project joins three products. The first is an open set of design and delivery tools: city planning, engineering models, quantities, prices, manufacturing plans and maintenance records. The second is the railway and its industrial system: trains, stations, guideways, workshops, tools and repair methods. The third is open control software, researched for train protection, supervision and faults. The software can be developed and tested before there is a railway; it cannot authorize movement of an unqualified train.

Local capability is part of the design

Local manufacture is more than a final assembly address. A workshop needs drawings, materials, process knowledge, inspection methods, fixtures, replacement parts and the authority to revise a design. Concrete precasting, structural fabrication, interiors, wiring, winding, assembly and routine maintenance may offer different opportunities in different places. Specialist bearings, electronic devices, cells or safety equipment may still come from outside the country. The origin and value added of each item should be recorded honestly.

Repairability changes engineering choices from the start. A removable panel needs accessible fasteners and space for a tool. A door mechanism needs a route out of the car. A bogie needs lifting points, service clearances and a realistic replacement sequence. A beam or bearing must be inspectable after installation. These details may add a little material or a manufacturing step, yet reduce years of downtime and dependence on one supplier.



Concept of a local vehicle assembly workshop

Concept illustration. The body, bogies, doors, glazing and interior appear as separable assemblies; their actual workshop methods and tolerances require controlled design work.

Compare complete systems

Transport concepts are often compared using one attractive number: a vehicle weight, a top speed or the price of a beam. A useful decision needs a common passenger service and a complete system boundary. A light vehicle may require more expensive propulsion equipment. A shorter station may need more trains. A low initial price may increase energy, maintenance or renewal costs. The design tools therefore carry the same candidate through vehicle, civil, electrical, service and economic accounts before ranking alternatives.

Some inputs can be calculated early, while others require a survey, a supplier or a physical test. The early estimate should show that difference. Unknown ground conditions do not become free foundations; unlocated equipment does not become a zero-mass vehicle. This makes the design more useful to a city even when it is incomplete, because the next decision and the cost of uncertainty are visible.

Open design knowledge also makes independent challenge possible. Another engineer can inspect an assumption, repeat a calculation, propose a different material or test a different train. A failed option remains valuable if it shows which constraint matters. OpenSourceRail's long-term value is therefore not one claimed winning configuration. It is a method for creating, checking and improving practical railway choices while retaining public knowledge of why each choice was made.

1.3 The passenger journey

A railway starts as a promise to people: reach the station, enter it, board safely, travel comfortably, make a connection and leave without unreasonable delay. The physical design should be judged against that journey. A drawing of a fast train says little about walking distance, crowding, step-free access or what happens when the service is disrupted.

Access before speed

Station spacing changes the balance between access and travel time. More stops can shorten the walk but add dwell, braking and acceleration. Fewer stops can speed a train while moving the benefit away from people who need it. A city proposal must therefore consider housing, work, schools, public space, transfers, barriers and the safe routes between them. The exact station point also affects utilities, property, foundations, entrances and rescue access.

The six-car reference places two doors on one side of each 17-metre car. It plans 44 fixed seats and two wheelchair places per car, with no planned standing. Across the train that is 264 fixed seats and 12 wheelchair places. The seating policy is a comfort decision, not permission to ignore crowd, evacuation or accidental-load cases. One retained structural control considers 160 people per car; it remains a separate load case even though routine service is planned for a smaller occupancy.



Passenger cabin concept with seating and a wheelchair space

Concept illustration. Door positions, boarding gaps, wheelchair circulation, restraints, fire performance and all installed clearances still need measured and controlled engineering.

A service is more than a timetable

The familiar two-minute interval is a named comparison case. With 276 planned places per train it yields 8,280 places per hour in one direction by arithmetic. It does not establish that train protection, dwell, terminal turnback, power or fleet availability can deliver such a service. A different formation or headway could be better for a specific city. An earlier 120-person train and 14,400-place comparison remains available for research, but its occupancy and service assumptions cannot be substituted for those of the current reference.

A credible timetable needs the route geometry, gradients and curves; the available traction force rather than only the required force; independent braking under adverse adhesion; station dwell and platform exchange; protected

separation; terminal operation; and recovery from a failed train. Energy and crew requirements follow from the same timetable. OpenSourceRail retains a 72 km/h launch cap in one stopped-network proposal, but the final speed is a candidate and corridor decision.

Ordinary days and difficult days

Passenger protection is most visible when a system fails. The railway must consider a train stopped between stations, a door fault, loss of wayside power, a hot day, a wet rail, a closed platform or a blocked evacuation route. Service batteries support lights, ventilation, communication, doors and protection where required; they are not traction batteries in the reference train. Independent brakes and rescue arrangements remain necessary even if a station propulsion system can recover energy in normal operation.

Accessibility reaches beyond the train floor. It includes station entrance, lift or ramp reliability, the platform and train interface, wayfinding, wheelchair and mobility-device space, sensory access and assisted evacuation. Each proposed station and vehicle change should preserve that complete path. The images in this chapter show an intention to make travel comfortable and inclusive. The actual result will depend on tested geometry, operations and feedback from the people who use it.

1.4 A reference, with room to explore

The development reference gives the project a common starting point: six 17-metre cars on passive two-axle steel-wheel bogies, running on conventional rails carried by concrete. The vehicle has no normal-service traction motors, traction batteries or levitation magnets. A passive aluminium secondary is retained as an interface for comparing station linear-motor propulsion. It is an interface choice, not a decision that linear motors have been installed.

This starting point simplifies some questions. A wheel-supported train can use familiar rail and running-gear principles. Removing normal-service traction equipment from the vehicle may make a car simpler to maintain, but puts more work into station equipment and its interfaces. Those system effects must be counted. Service batteries, independent brakes, protection, HVAC and rescue equipment still belong on or around the train.



Illustrated wayside linear-motor comparison

Concept cutaway. Wheels carry the vehicle; a passive fin sits between station-side motor primaries. Motor geometry, powered length and repeated-duty performance are not selected by this image.

Change the whole candidate

Research is permitted to challenge the reference. Powered wheels, onboard recovery, different station drives, alternate bogies, shorter cars, different formations and civil sections can all be explored. A change is meaningful only if its consequences are followed through the complete railway. A powered bogie, for example, adds motors, gearing, converters, cooling, wiring, protection, mass, maintenance tasks and perhaps different axle loads. It may remove or reduce wayside propulsion equipment. A fair comparison accounts for both changes and provides the same passenger service.

The project distinguishes the physical candidate from an evaluation of that candidate. If the geometry is unchanged but a new material property or magnetic-field hypothesis is tested, the physical train has not changed; the calculation has. A selection then records why one candidate is preferred for a particular purpose. These distinctions keep a favourable result from one model from being attached to another train without an explicit new evaluation.

Design question	What must travel with an alternative
Vehicle formation	Platform length, seats, fleet, mass, coupling, rescue and construction effects

Design question	What must travel with an alternative
Propulsion	Installed force, power, thermal duty, equipment, protection and maintenance
Running gear	Axle loads, wear, noise, bearings, brakes, access and inspection
Civil form	Load paths, bearings, foundations, erection, access and renewal
Service	Demand, dwell, frequency, energy, operating staff and spare fleet

There is no mandatory 900-metre or 100-metre powered installation. Those lengths appear in retained comparisons, and future designs can choose other arrangements. Nor does a short journey calculated from a prescribed acceleration prove that a motor can supply it. The railway must close force, current, heat, deformation and clearance together over actual operating conditions.

The older permanent-magnet maglev work remains an important historical research family, with its own source records and lessons. It is not the wheel-supported development reference. Preserving alternatives makes the project more open; keeping their evidence separate makes that openness technically credible.

1.5 Capacity, frequency and useful service

A railway can carry many people on paper and still offer an awkward journey. The passenger does not experience a capacity figure; they experience the walk to the station, the wait, the ability to board, the time on board and the certainty of reaching their destination. Capacity has to be tested against those experiences, and against what the railway can deliver reliably on an ordinary hot afternoon, not only under a perfect timetable.

Places and trains are different measures

The six-car reference has 44 fixed seats and two wheelchair places in each car. Across the formation that is 264 fixed seats and twelve wheelchair places, or 276 named passenger places. At one train every two minutes, simple arithmetic gives 30 trains per hour and 8,280 places per hour per direction. This is a useful reference control for comparing concepts. It is not an operating capacity until station dwell, safe train separation, terminal turnback, boarding patterns, availability and passenger demand have been shown to support it together.

The project retains a 120-person-per-train comparison that yields 14,400 places per hour at the same two-minute interval. That is an alternative formation and accommodation question, not a property of the seated reference train. The 160-person-per-car crowd load is a structural control. It asks whether the body can withstand a demanding occupancy case; it does not tell the operator to plan standing passengers in everyday service. Keeping these three quantities distinct prevents a strong beam calculation from becoming an unintended passenger policy.

The bottleneck moves

A shorter headway can increase offered places, but it also makes dwell variation more important. A wheelchair boarding event, a busy interchange or a door fault can hold the following train unless the stations, control system and recovery plan have enough margin. More trains require more vehicles, more power, more inspections and often more depot capacity. A longer train may relieve crowding but lengthens platforms, increases station and structure loads, and can complicate access on a tight site.

The retained-stop reference uses a 72 km/h launch cap. It is a design control for the journey calculation, not a promise that every alignment can reach that speed. Acceleration, braking, gradients and closely spaced stops can dominate travel time. So can a station placed on the wrong side of a road. A useful service comparison therefore holds the actual corridor and passenger objective constant while varying train length, frequency or propulsion. Otherwise a cheap vehicle can appear attractive only because it offers fewer journeys or serves fewer people.

Decision	Passenger effect	Engineering consequence
Add a station	Shorter access for some riders; longer ride for others	Platform, civil work, dwell, power and fleet may change
Run more often	Shorter average wait and more offered places	More trains, energy, staff and protected control capacity
Use a longer train	More places per departure	Longer platforms and different structural loads
Change stopping pattern	Different travel and transfer times	New timetable, capacity and recovery analysis

Reliability is part of capacity

An offered timetable is only useful when the fleet and infrastructure can maintain it. The model should include spare trains, inspection time, maintenance access, station equipment faults, power outages and the time needed to rescue a stopped train. A failure that blocks a single guideway can turn a high theoretical capacity into a long queue. The cost of that queue matters to a city's decision even if it does not appear on the construction bill.

OpenSourceRail's direction is to compare complete service packages. A candidate needs a route, station pattern, train and propulsion arrangement, protected separation concept, maintenance plan and cost range. The current tools can connect many of these assumptions and expose what changes after an edit. The passenger demand, safety margins and failure rates needed for an operating timetable still require local evidence and independent

review.

II. From city to corridor

Planning, service and a worked station revision.

2.1 From a city brief to an alignment

A city does not need an identical railway merely because another city has a promising train. Population, topography, heat, flood risk, streets, utility networks, land rights, transfers and local production capacity change the answer. The planning tools begin with geography and a transport need, then make each route and station proposal explicit. The objective is a comparison that a city can question and revise, rather than a map whose apparent precision hides missing surveys.

Start with people and places

An initial brief describes where people travel, when they travel, what connections they need and which journeys are currently difficult. It should include walking and wheelchair access, climate, travel time, comfort, safety and an affordability range. Many of these inputs are incomplete early on. A useful outline can still be made if the missing demand, prices or site facts are labelled as assumptions with a clear plan to improve them.

The city catalogue brings together route lines, stop and transfer markers, street and place context, and other available geographic layers. A planner can propose an alignment, move a stop, alter a dwell assumption or change the service. The revised proposal is then evaluated with a declared vehicle and propulsion policy. Its fleet, energy, civil quantities, cost and dossier can be compared with the parent proposal. An edit to one station need not recalculate an unchanged carbody; a change in a common material price may affect many cost accounts while leaving the geography alone.

Early question	Evidence needed before a build decision
Where should the line go?	Surveyed terrain, land, utilities, structures and local access
Where should a station sit?	Walking routes, platform fit, transfers, foundations and evacuation
How often should trains run?	Demand, dwell, protected separation, terminals and spare fleet
What does it cost?	Candidate quantities, local rates, site works, risk and renewals

Maps are a starting point

A visible route is not yet a buildable alignment. Railway horizontal and vertical geometry, gradients, drainage, bridge interfaces, foundations, access roads and temporary construction space have to be checked. A station marker may be close to a desired destination but difficult to reach across a major road. A neat viaduct line may intersect a utility or require costly ground treatment. The Workbench can identify such questions and show which results depend on them; it cannot create a survey that has not happened.

Native GIS editing is being connected to this process. An editor can export a city project, change a station and return the edit for recalculation. The Samawah and Baghdad development journeys have exercised two consecutive native edits each, including repricing and a refreshed project. Those tests show that the software can carry an edit through a defined workflow. The survey accuracy and engineering acceptability of the route remain separate.

Keep the proposal attached to its assumptions

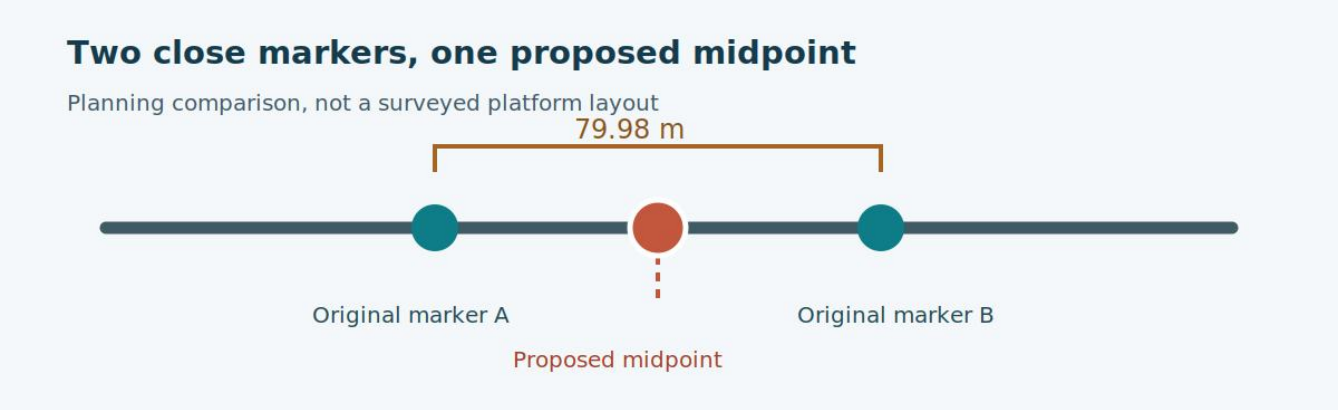
Each corridor revision should retain the route geometry, stops, chosen train, service assumptions and price basis that produced its result. This prevents a later design from borrowing an attractive cost or journey time from a different candidate. It also lets a reader see which inputs changed when two outcomes differ. If the edit reduces one platform or changes one dwell, the report should show the effect on station count, travel time, trainsets, energy, civil scope, capital cost and whole-life cost. A zero change is informative when it is verified; it should not be silently omitted.

The best first result is therefore a questionable, traceable outline. It should be detailed enough to expose the major trade-offs, honest about missing site evidence, and easy to revise when a city supplies better information.

2.2 Samawah: one edit through the system

Samawah provides a concrete example of why railway decisions must remain connected. Two raw station markers on the selected line are about 79.98 metres apart, while the planning case assumes a 120-metre platform. A station merge was proposed to resolve the apparent conflict. The important question is not whether one marker can be deleted from a map. It is what the edit changes for passengers, vehicles, structures and cost, and what it still cannot establish.

The v0.7 comparison replaces the two markers with one midpoint marker without changing the line's route coordinates. The published example had already grouped the pair into one operational station complex. The raw marker count therefore falls from 26 to 25, while the operational count remains 25. The route stays about 57.87 km and the modelled fleet stays at 81 trainsets. Neither outcome means the platforms or walking routes have been surveyed.



Schematic of the two nearby Samawah markers and proposed midpoint

Schematic only. The separation is from the planning record; platform and pedestrian geometry have not been surveyed or approved.

Measure in the planning comparison	Before	Midpoint proposal
Raw station markers	26	25
Operational complexes	25	25
Modelled trainsets	81	81
Base installed capital	about $1.141491 billion	about $1.141481 billion
Base whole-life present cost	about $2.258426 billion	about $2.258656 billion

The model's installed capital falls by roughly $10,000, yet its whole-life present cost rises by roughly $230,000. The selected line's one-way time grows by about 1.5 seconds. Most of the whole-life increase comes from the energy account. The access allowance also changes: one notional repair length is replaced by a shorter midpoint access hypothesis with a separate risk allowance. Those figures are useful because they show the direction and scope of the model response. They are not a local construction quotation, a walking survey or proof that the station is feasible.

What stayed the same matters too

The other four lines were not redrawn. The vehicle and manufacturing reference did not change. The civil quantity identity was preserved because this particular edit did not revise the modelled guideway. All fourteen cost scopes were compared, including scopes with zero change. Reusing unaffected results keeps the study focused and makes it easier to understand why a cost moved.

The comparison also keeps its adverse findings. A line-1 coast-reach question, actual propulsion and braking evidence, transferred prices, platform fit, catchment and walking access still need work. A separate compact-drive research case once examined a similar midpoint idea with a different body candidate. Its fleet or cost result is not transferred into this passive-wheel comparison.

The next on-site decision

The midpoint should be tested against the real pedestrian network, frontage, crossings, utilities, station entrance and platform geometry. A suitable station might be placed at the midpoint, at one original marker or elsewhere; the model alone does not decide. The value of the software journey is that each new proposal can be carried through the same service and cost questions, while the unresolved site evidence stays visible to the people making the decision.

2.3 Networks, service and city variety

OpenSourceRail contains planning outlines for 266 cities. This breadth is useful for exploring how different urban patterns might influence a railway. It is not a claim that 266 independent surveyed, engineered and priced systems are ready to build. Many city outlines use shared assumptions or transferred rates. A city dossier can help an owner ask better questions while remaining clear about the evidence that is still local and absent.

One city, several possible services

An alignment is only part of a network. The number and position of stations, interchanges, service hours, frequency, train formation and spare fleet affect the usefulness and cost of the railway. A dense centre may reward shorter walks and frequent stops; a longer peripheral corridor may need a different spacing or an express pattern. The same train may not be the lowest-cost choice for both. Passenger demand by time of day matters as much as a total city population.

The software can model station counts, route lengths, timetables, fleet and energy under stated assumptions. It can compare low, base and high cost cases across capital, operating, renewal and financing accounts. These are design stage scenarios. They cannot replace observed demand, protected headway, supplier prices, actual ground conditions or approved operations. Where a route is drawn through uncertain terrain, the corresponding civil and risk cost should remain provisional.

Geography and economics must move together

A change in route may alter length, curves, gradients, bridges, station access and land take. That can change travel time and energy as well as structures and cost. A change in demand can alter the fleet and power requirement even when the line stays in place. Conversely, a change in local wages should reprice the applicable labour work without pretending the railway geometry changed. This is why the planning platform keeps a relationship between inputs and results, instead of publishing a new opaque total after every edit.

Baghdad is a useful second test after Samawah because its corridor and city context differ. A native GIS workflow has returned a sequence of station edits from both cities to the calculation and back to a refreshed project. That checks the continuity of the software route. The engineering case still needs its own surveyed alignment, site-specific structures, station access, demand and candidate-specific operating assumptions.

From outline to city decision

The first decision might be which corridor deserves a survey, where an interchange should be studied, or which train and civil family merit a detailed comparison. It need not be a commitment to construct. A strong city proposal states its transport objective and alternatives, records the inputs, shows the cost and service sensitivities, and lists the missing evidence likely to change the ranking. It also includes the people who will use the system: wheelchair users, older passengers, children, shift workers and those whose journey crosses the proposed line rather than following it.

The 266-city catalogue is a broad planning library. The detailed v0.7 work concentrates on one Samawah corridor, then one Baghdad corridor, so that a complete change can be followed through design, cost and delivery before the same method is offered more widely.

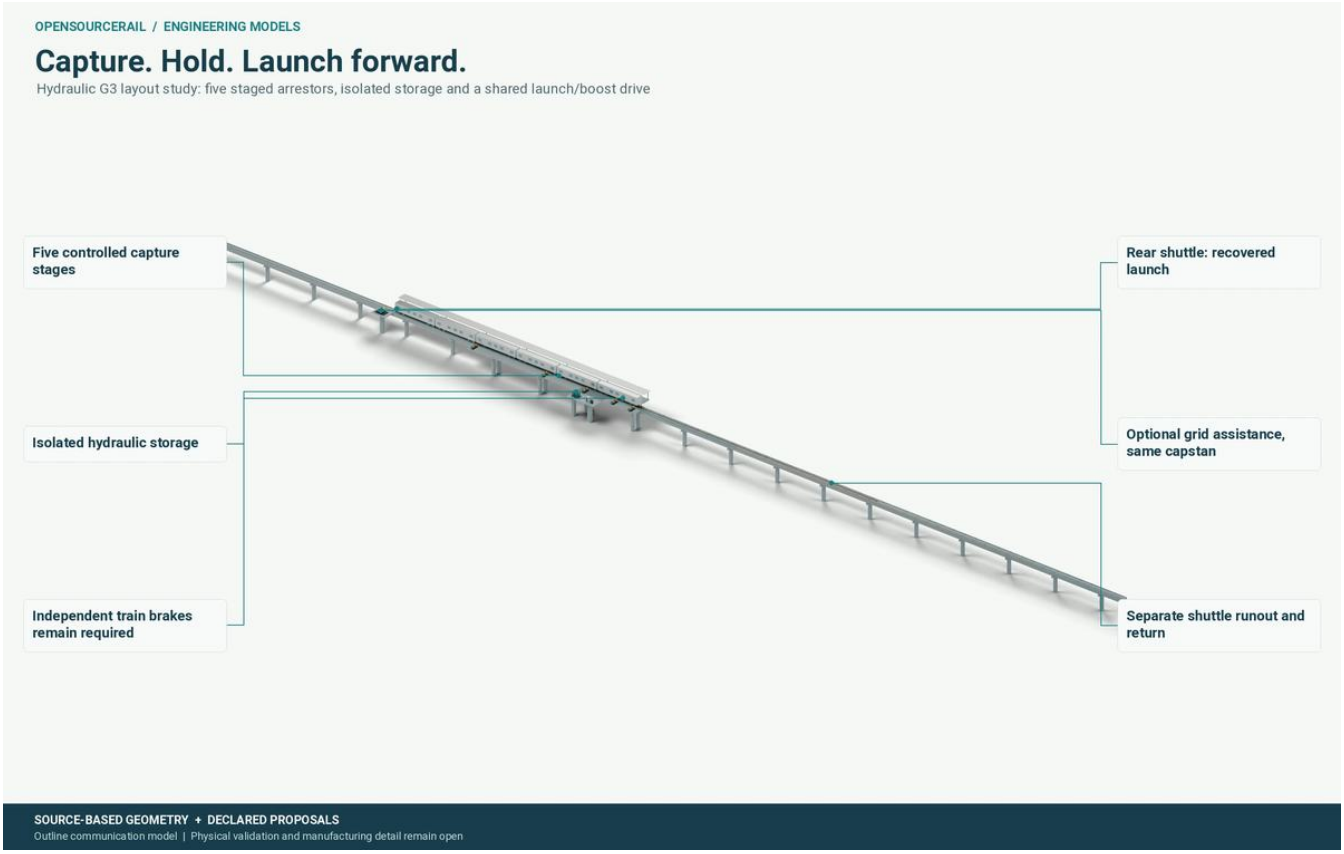
2.4 Stations and the streets around them

A station is a connection between the railway and the city, not simply a point on a line. Its success depends on whether a person can find it, reach it safely, enter without unnecessary effort, wait comfortably and transfer to the next part of the journey. The arrangement must also give staff safe access to the railway, its power equipment and its structure.

The journey begins outside the platform

An elevated line can save street space, but height creates access work. Entrances, lifts, stairs, ramps where feasible, wayfinding and emergency routes have to connect both sides of any road barrier. A station placed a short distance from a market may still be a poor choice if the legal and accessible walking route takes a long detour. Heat, shade, lighting, drainage, crossing safety and the availability of local transport belong in the station brief from the start.

Platform length must follow the selected train and its stopping tolerance. The current reference formation is roughly 104.5 metres long under the proposed inter-car gap assumption; a 120-metre platform is the planning control used in the Samawah comparison. That allowance cannot be treated as a complete platform design. End clearances, doors, accessible boarding positions, sightlines, egress, structures and future operating choices must still be resolved. Changing the formation after setting the station layout would change much more than a vehicle drawing.



Station and street concept

The concept shows a connection between rail and street. It does not settle the site's land, access, fire or structural design.

A stop and a station complex are not the same thing

City source data may contain several markers for what people understand as one destination. In Samawah, two markers were found about 80 metres apart, closer than the platform planning assumption. Their merger reduces the raw marker count from 26 to 25 while the operational station-complex count remains 25. That distinction matters: the data cleanup can be useful even when a model says that the number of built stations is unchanged.

The midpoint of two markers is a comparison device, not automatically the best entrance location. One side may have better walking access; the other may have poorer ground or a utility corridor. A real station choice requires an accessible-catchment study, land and ground checks, utility information, public-realm design and engagement with people who will use it. The calculated capital and travel-time changes in the Samawah chapter help frame that inquiry; they cannot decide it alone.

The station as a maintained asset

Platforms, lifts, lighting, fire systems, drains, signs, communications and power cabinets need inspection and replacement space. A design that can be built only by closing a major street for months, or maintained only by blocking both tracks, may cost more over its life than a slightly larger initial structure. Access for cleaning, repair vehicles, emergency services and safe isolation should be tested before the layout is fixed.

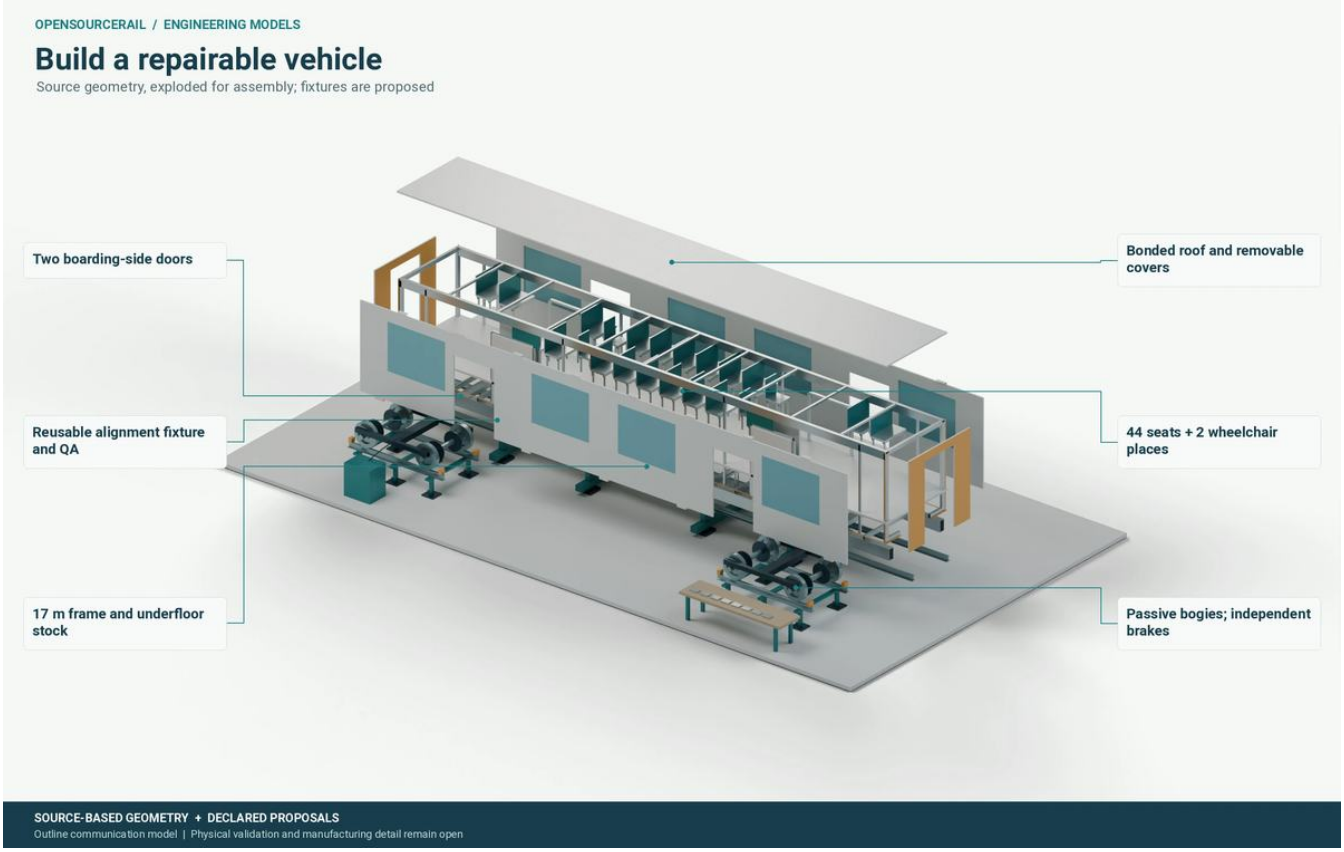
A station proposal can move through increasing levels of evidence. First comes the city outline and the passenger purpose. Then come surveyed geometry, access routes, ground and utilities, a matched train and service, and a civil concept. Detailed fire, egress, structural, electrical and operational reviews follow. The Workbench's value is to retain those links and show what a station edit changes, while displaying missing evidence plainly. It should never make an unsurveyed map look like an approved site.

III. The physical railway

Vehicle, passenger interfaces, power, stopping and civil works.

3.1 Vehicle and running gear

The reference train has six 17-metre cars. A passive two-axle bogie supports each end of each car, giving twelve bogies, 24 wheelsets and 48 axleboxes for the formation. Steel wheels run on conventional rails carried by the concrete guideway. Motors and gearboxes are absent from this reference running gear; springs, damping, bearings, brakes, monitoring and emergency support still have to be installed, inspected and maintained.



Source-based vehicle assembly view

Engineering communication model. It separates major assemblies and shows their proposed relationship; it is not a complete installed-mass or approved manufacturing drawing.

Design the car as an installed system

A carbody is not just a shell. Its floor, sides, roof and joints must carry passengers, equipment, bogie reactions and accident loads while leaving room for doors, windows, ventilation, cables, batteries, seating and maintenance access. A material can appear light in a small structural test and still require heavy joints or reinforcement around openings. The project therefore compares body families with their complete load paths rather than treating a single beam stiffness result as a qualified car.

The proposed manufacturing breakdown names 605 leaf-part types in 40 assemblies across the six-car train. It counts major structural stock, wheelsets, bearings, doors, seats, batteries, HVAC, lighting, wiring, service equipment and many fasteners. This is a substantial outline inventory, but the list is not a measured installed train. Gross stock and assumed bought equipment masses must be reconciled with actual fitted parts, centre of gravity, inertia and available space.

One detailed proposal estimates 78.49 to 106.60 tonnes across the train from stock and component allowances, with 89.81 tonnes as its base subtotal. These are bounds within that particular proposal, not an approved tare or proof of a weight saving over another candidate. Unlocated equipment, overlapping reserves and supplier internals must be resolved before an installed mass can be stated. Those quantities matter to axle load, braking, energy, beam design and fleet cost, so an optimistic assumption cannot remain isolated in the vehicle chapter.

Formation and interfaces

The six cars need five inter-car mechanical couplers and separate passenger gangways. A gangway gives people a route between cars; it does not replace a load-bearing coupling or an emergency rescue connection. Air, signals, lighting and other essential services also cross the joint. Their bending, movement, fatigue and removal paths have to fit the operating curves and the workshop process.

A revised manufacturing proposal increases the inter-car gap from 500 to 650 millimetres, taking its body-plus-gap envelope from 104.50 to 105.25 metres. The extra space may ease certain interfaces, but platform margins, coupler projection and stopping tolerance must be checked on the same candidate. A formation change cannot quietly inherit the parent platform or fleet result.

The bogies themselves present a similar trade-off. A familiar wheel-and-rail interface offers an established body of engineering knowledge, but wheel wear, rail contact, noise, bearing service, wheel-slide protection and wet-weather braking remain real tasks. The project needs dynamic and physical checks, not only a visual fit of wheels beneath a CAD body.

3.2 Inside the vehicle

Passengers see a floor, seats, windows and doors. Beneath those surfaces is an arrangement that must carry people safely and let maintainers reach the equipment. The body, bogies, batteries, ventilation, wiring, couplers and interior are one installed vehicle. Their combined geometry and mass decide what the track and structures experience, how the train stops and how easily it can be repaired.

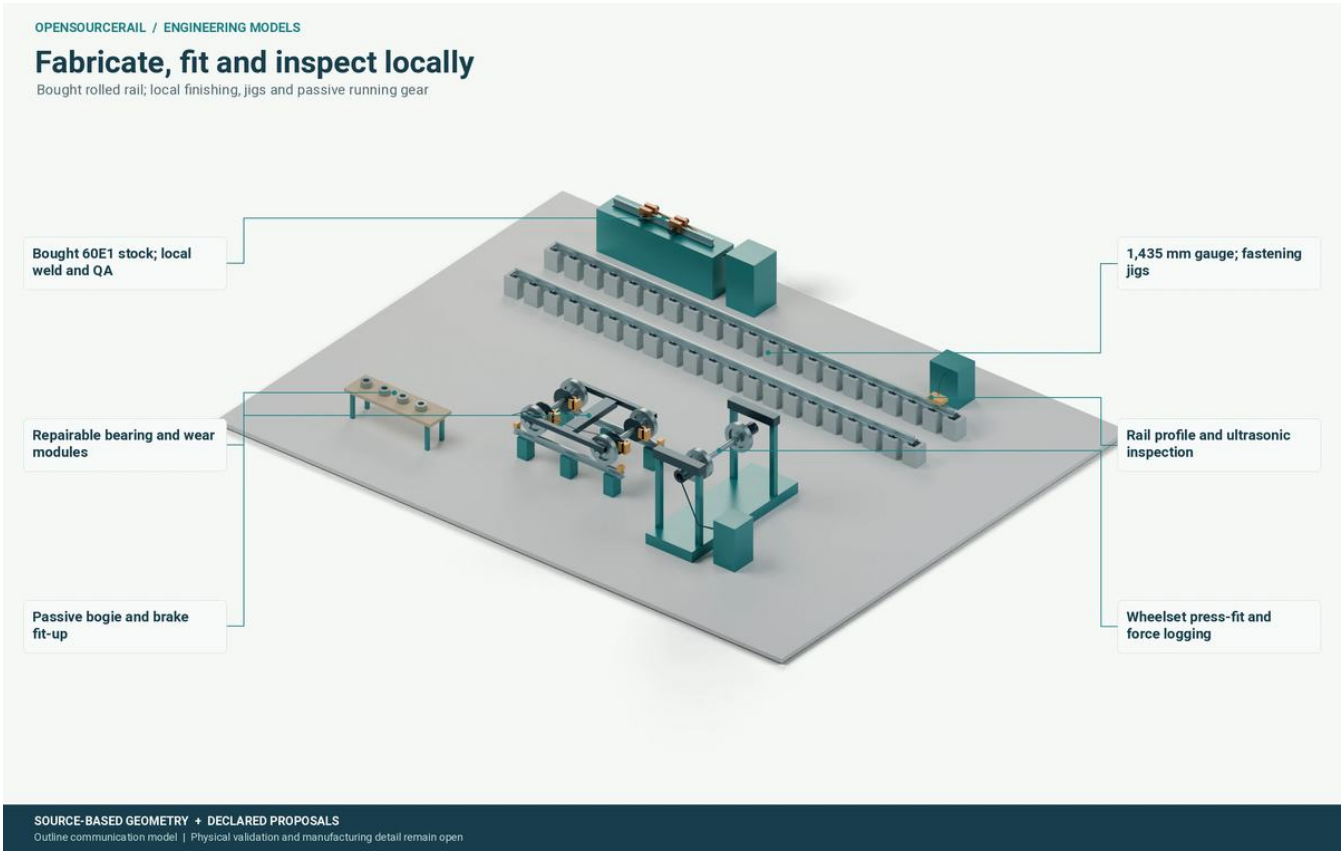
The body is a load path with openings

A railway car must transfer vertical, longitudinal and lateral forces among floor, sides, roof, bogie connections and couplers. Doors and windows remove material from the sides just where a simple continuous beam model might expect it. The remaining structure may need frames, skins, joints and local reinforcement. Seats and equipment also impose attachment loads, while thermal movement and fatigue test the joints over time.

OpenSourceRail explores different structural grammars and materials rather than fixing every member before analysis. A braced frame, a bonded shell and a rigid jointed body cannot be judged by exactly the same simplified mechanism test. Their manufacturing routes also differ. A lighter panel may need more costly curing or a more demanding repair process; a familiar metal route may be easier to inspect and source locally. The appropriate comparison includes installed stock, joining, fire response, fatigue, corrosion and replacement, not a single material's strength-to-weight ratio.

The bogie carries more than weight

For the reference six-car train, twelve passive bogies and 24 wheelsets support the formation. They guide the wheels along the rail, transmit braking and lateral forces, and isolate the body from track irregularities. Bearings, springs, dampers, brake components, sensors and safety restraints need their own mounting and access. Wheel profile and rail contact affect wear, noise and adhesion. A body that fits over a bogie in a static drawing may still foul it on a curve, during suspension movement or when a wheel is worn.



Rail and running gear study

An engineering model of the running interface. Dynamic clearance, wear, braking and maintenance need their own checks.

An alternative powered-wheel bogie may simplify station-side propulsion while adding motors, gearing, converters, cooling and protection beneath each car. That changes unsprung or sprung mass, clearances, energy, braking and workshop tasks. The passive reference and the powered alternative should therefore have separate, complete vehicle and service results.

Space is an engineering resource

Seats, wheelchair places and circulation compete with equipment cupboards, ducts, structural members and emergency routes. The same underfloor region may be proposed for batteries, cables and removal access. A large unit that fits in its installed position may still be impossible to take out without removing a bogie or blocking the platform. A useful three-dimensional model shows both the operating envelope and the maintenance envelope.

The current parts proposal names 605 leaf-part types in 40 assemblies and over 110,000 expanded pieces across a formation. These counts expose the many connections behind a simple visual concept. They are not a released manufacturing bill: some masses are allowances, some fit positions remain generic and some supplier internals are not fully counted. A complete installed mass must reconcile the physical item, its quantity, location, fixings, protection and replacement route. The low, base and high stock allowances are useful for sensitivity, but the selected vehicle must be weighed and checked at the relevant build stages.

The interior deserves its own tests

An accessible passenger place is more than a marked rectangle. Boarding gap and step, turning area, securement, handholds, reach, emergency egress, lighting and communication all matter. The one-sided two-door arrangement must suit the chosen platform side and station layout. A crowded structural test does not set the normal seating policy, but it does remind the team to consider misuse, unusual loads and evacuation. Passenger feedback and physical mock-ups can reveal problems that a detailed CAD model misses.

3.3 Doors, cabin and access

The proposed train has twelve one-sided passenger door openings, two per car. Doors are a small part of the vehicle's visible area but a large part of its daily reliability and passenger protection. They must open and close without striking the body or platform, detect an obstruction, remain secure in motion, allow emergency release and be repairable without dismantling the car.

One opening as a representative module

The current door study counts 74 mechanical and electrical part types and 694 individual pieces for one opening. The make-or-buy proposal assigns 97 pieces to local fabrication, 568 to standard purchase and 29 to purchase subject to qualification. It includes the header, rails, carriage, belt and pulley, drive, leaves, glazing, sensors, seals, wiring, fixings and inspection items. These counts help expose the work hidden by a simple animation of two moving leaves. They do not mean the door is ready for manufacture.

The next design step is to establish mating datums: where the header attaches to the body, how the rail and carriage align, how the belt and drive meet, where sensors sit, and how the assembly can be lifted out. Several proposed operator parts still occupy a generic central header zone instead of fully resolved fitted positions. A two-metre removal sketch does not represent the full 2.9-metre guide stock. These are specific reasons to keep installation and maintenance open, even though the inventory is detailed.

A cabin that works in service

Each car's 44 fixed seats and two wheelchair places provide a clear normal accommodation policy. Their location must still allow boarding, circulation, evacuation, cleaning and access to service equipment. Windows, seat fixings, handholds, lighting, ventilation and interior panels must meet their own fire, impact and maintainability requirements. A bright concept interior is an invitation to test the layout with passengers, not a substitute for those tests.



Passenger interior concept

The accessible interior is a design intention. Actual circulation, doorway, platform gap and emergency arrangements remain engineering tasks.

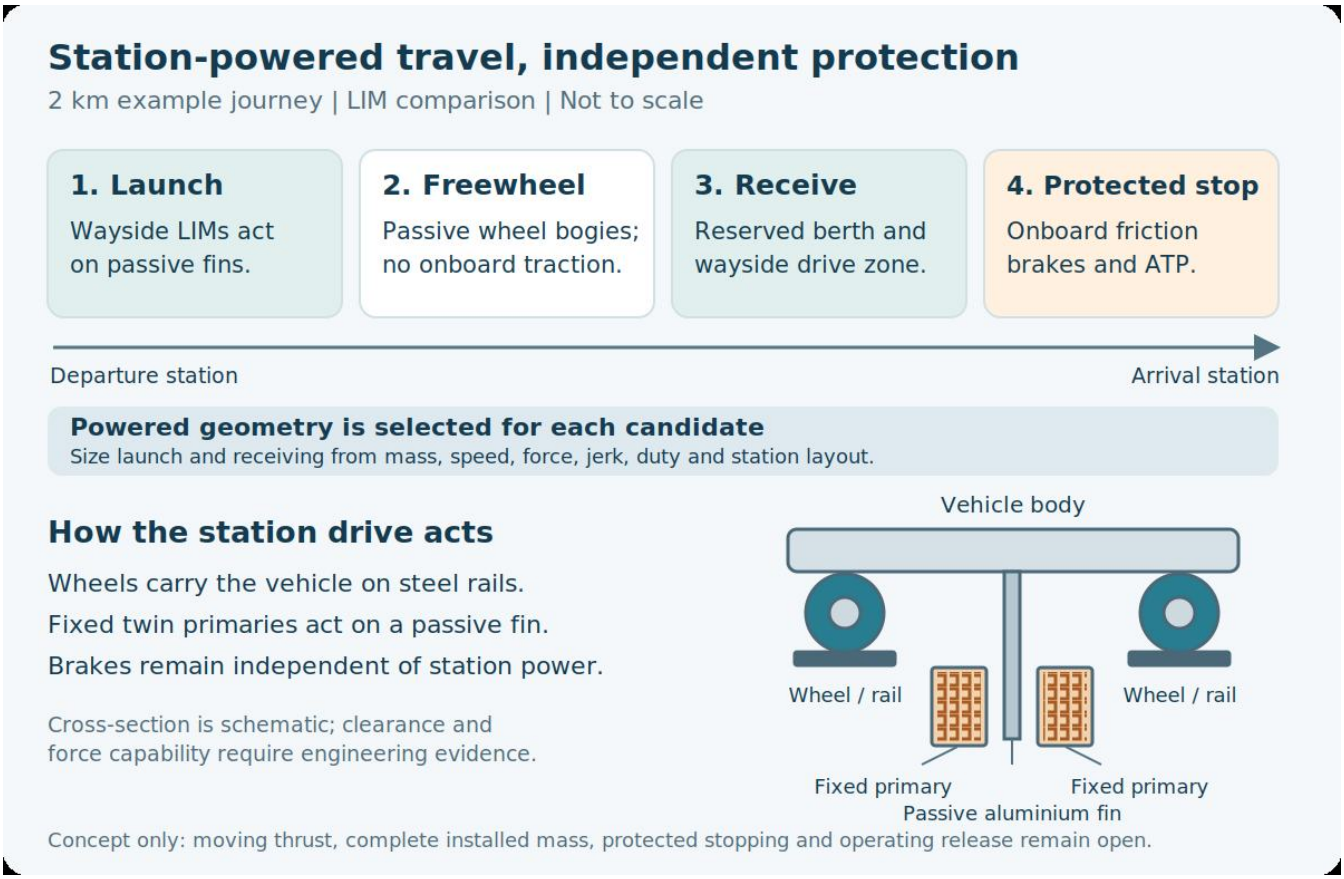
The lighting proposal uses replaceable LED modules in two rows per car, with separate protected circuits and independent emergency fixtures. The counted normal lighting demand is modest relative to propulsion, but the

design must still establish illumination, glare, electrical protection, electromagnetic compatibility and hot-weather performance. The same principle applies to HVAC: a catalogue load or a CAD envelope is a starting point, while real airflow, pressure, temperature and service access must be verified.

The door operator is chosen as the v0.7 representative manufacturing and maintenance module because it forces mechanical, electrical, body and passenger questions to meet. A controlled package should show dimensions, tolerances, materials, assembly sequence, tooling, QA limits, removal path and replacement record for one exact vehicle candidate. A business-system record can then follow the physical module through receipt, inspection, fitting, service and eventual replacement.

3.4 Propulsion, power and braking

The wheel-supported reference separates carrying the train from driving it. Wheels and rails bear the load. Station-side equipment is being compared for launch and arrival, with a passive aluminium interface on the vehicle for linear-motor research. Other candidates may use mechanical launch equipment, powered wheels or different recovery arrangements. None is selected as an installed system merely because a schematic looks simple.



Launch, freewheel, receive and protected stop concept

The diagram explains one operating idea. The actual force, powered length, energy transfer and protected braking distance depend on the chosen candidate and route.

What a station drive must achieve

A station-powered train could receive its main acceleration near departure, coast through much of the interstation distance and be received at the next station. This may reduce train-borne traction hardware. It also concentrates power and thermal duty at stations and introduces equipment gaps between stations. On an uphill route, a coasting train may not reach its destination within the proposed speed limits. On a downhill route, it may need additional speed control. A failed station or an unexpectedly stopped train makes rescue and independent stopping essential.

A linear motor needs an attainable force map across speed, current, voltage, air gap and temperature. A model that matches an ideal travelling field is a useful method check, but it does not include every slot, end effect, inverter or moving support of a manufactured motor. The passive fin, its mounting, thermal expansion and wheel/rail movement must fit the same clearance envelope. Powered length is consequently a result of design, not a fixed 100-metre or 900-metre prescription.

Mechanical capture and release is another research family. It asks whether a station can recover some arriving kinetic energy, store it briefly and help launch the next train. The physical mechanism would need contact geometry, energy storage, losses, guards, fault handling, maintenance and a reset sequence. Energy recovered in a calculation does not automatically become available to the next departure. Powered-wheel alternatives must

likewise carry their added motors, gearboxes, converters, cooling and bogie changes.

Independent stopping and service power

The reference train requires onboard service, emergency and parking brakes that do not depend on station power. Stopping must be checked at low adhesion, on gradients, under crowd loading and after relevant faults. One retained adverse calculation from 20 m/s uses a low adhesion assumption and a downhill grade; it gives a stopping distance of roughly 700 metres including response. That is a sensitivity, not a validated wet-rail brake curve or an approved separation distance. Wheel-slide protection, thermal duty and degraded operation need their own evidence.

Service batteries support HVAC, lighting, doors, communications and protection. A retained 82.8-kWh-per-car pack is a sizing hypothesis, not a qualified installation. Heat can dominate the design: a cell with a 45°C ambient limit cannot be kept below that limit in 50°C surroundings by a radiator alone. Cooling, enclosure, charging, emergency endurance, fire isolation and replacement access must be designed and tested together.

The main question is not which drive sounds most elegant. It is which candidate supplies the required service with credible force, safe stopping, bounded heat, repairable equipment and the lowest justified whole-life cost.

3.5 Stopping, faults and rescue

The safest journey is one whose unusual moments have been designed as carefully as its normal motion. A train can lose a launch, stop between stations, encounter poor adhesion, meet an occupied route or need to be evacuated in heat. A proposed drive and timetable are credible only when the train can be brought to safety and the passengers cared for in these states.

Braking belongs to the train

Station-side equipment may help accelerate or receive a train, but the reference requires service, emergency and parking braking independent of that equipment. This gives the train a means to stop after a station power fault or missed receiving event. It does not remove the need to prove the brake's actual force and repeated thermal capacity. Wheel and rail adhesion varies with water, contamination and load; wheel-slide protection has to preserve steering and stopping rather than assume a fixed friction number.

Stopping distance includes perception or detection, control response, equipment build-up and actual deceleration. It depends on initial speed, gradient, mass, available adhesion and the faults considered. One retained downhill, low-adhesion calculation from 20 metres per second gives a distance around 700 metres when response is included. That is a warning about the size of the problem, not a validated minimum separation. The protected route and service headway must ultimately use the accepted worst relevant cases and approved margin.

A failed drive changes more than arrival time

If a station cannot launch a train, passengers may be held at a platform. If reception fails, the train needs a safe independent stopping location. If it stops between stations, the project must decide how it will be secured, communicated with, reached and moved. Gradients, curves, weather and the absence of a continuous powered rail can make recovery difficult. A candidate may need a rescue vehicle, a tow interface, local storage or another provision. Whichever path is chosen has mass, civil access, maintenance and operating costs.

Emergency planning also changes stations and guideways. An elevated track must have a way for responders to reach people and for people to leave a disabled train, including those with reduced mobility. The guideway needs working space and protection from adjacent traffic or power equipment. Communications, lighting, ventilation and battery endurance must last long enough for the chosen procedure. These are candidate requirements, not generic promises from a diagram of a train at a station.

Faults can combine

A hot day can increase cooling demand at the same time that a power system is stressed. A vehicle fault can lengthen dwell and disrupt the following train. A crowded carriage can change brake load during poor adhesion. Engineering should therefore test combinations that matter, while keeping the scenarios bounded and explainable. It is equally important to show what has not been tested. An apparently safe result under normal dry conditions cannot be borrowed for a wet, loaded, downhill fault.

The current platform has software controls and conditional journey models that can exercise parts of these scenarios. Actual propulsion availability, brake behaviour, thermal endurance and human evacuation need measurements, procedures and independent review. Recording that boundary is a practical part of safety: it tells the next engineer what evidence is still needed before a timetable or design can be accepted.

3.6 Heat, ventilation and electrical life

Power reaches a passenger railway through more than a propulsion device. Stations and vehicles need lighting, ventilation, doors, communications, controls, fire protection and safe supplies when the normal source is lost. Each system occupies space, adds mass and creates heat. Their combined installation can change the vehicle, structures and operating cost as much as a prominent motor choice.

Separate traction from essential services

The wheel-supported reference avoids onboard traction motors and traction batteries. This does not mean the cars have no electrical system. Door operation, emergency lighting, communication, control, ventilation and other essential functions need a protected supply and a defined response to faults. A retained service-battery hypothesis uses about 82.8 kWh per car. Its actual duty, mass, chemistry, cooling, isolation and replacement interval have to be sized against the selected equipment and emergency scenario. The number is a research input, not an installed procurement specification.

An onboard recovery or wheel-traction alternative remains possible. It would add different running gear, protection, charge or fuel logistics, mass, thermal loads and maintenance work. It must therefore be carried as a new candidate with its whole-system effects, rather than inserted into the reference train as a small component substitution.

Hot weather changes the design

Samawah and Baghdad make climate a central engineering input. Cabin comfort depends on passenger load, sunlight, glazing, insulation, ventilation, infiltration at doors and equipment heat. Electronics and battery modules can have narrower temperature limits than the cabin. A battery study at a 45°C module ambient does not demonstrate operation at a 50°C site condition without a cooling model and suitable measurements. Shading, airflow, equipment placement and service access should be assessed together.

Cooling has a cost on both sides of the ledger. A larger unit consumes energy, mass and roof or underfloor space. An undersized unit can shorten component life, cause service failures and increase replacement cost. A plausible heat balance must follow energy from its source through the equipment to the air or structure that removes it. The same discipline applies to station drive equipment, braking resistors, transformers and power electronics during repeated departures, not only during one short demonstration.

Current, force and heat must agree

A proposed station-powered linear drive can be attractive because it keeps major traction equipment off the train. Its available force depends on geometry, speed, current, magnetic field, temperature and the secondary conductor. Electrical input becomes motion, stored field energy and heat; the balance has to close for the actual operating duty. A calculated peak force at one position cannot by itself support a timetable.

The next practical evidence is a force-speed and thermal map with a bounded electrical supply, repeated launch cycles and measured or justified material behaviour. It should include losses and cooling at the drive, its structure and the vehicle interface. Braking remains independently sized, with stopping and rescue possible when the receiving drive is unavailable.

Design for isolation and replacement

The electric railway must tell maintainers which equipment can be safely isolated and replaced, and how a failed module is identified. Cable routing, connectors, water ingress, fire separation, access panels and working space are part of the design. A locally repairable system needs a parts list and inspection method that a local workshop can actually use. The Workbench can connect a failure record to its component and candidate, but the approved maintenance intervals and procedures must come from qualified equipment and operating experience.

3.7 Stations, structures and construction

The reference railway uses conventional rails carried by concrete structures. Mostly elevated, prefabricated construction is intended to reduce occupation of busy streets and to make repeatable local production possible. A structure still needs to be designed for its particular ground, water, utilities, access, loads and construction method. Repetition can lower cost only after the repeated unit itself is complete and safe.



Concept of a precast guideway installation

Concept illustration. The lift, crane, restraints, traffic management and temporary stability must be designed for each installation.

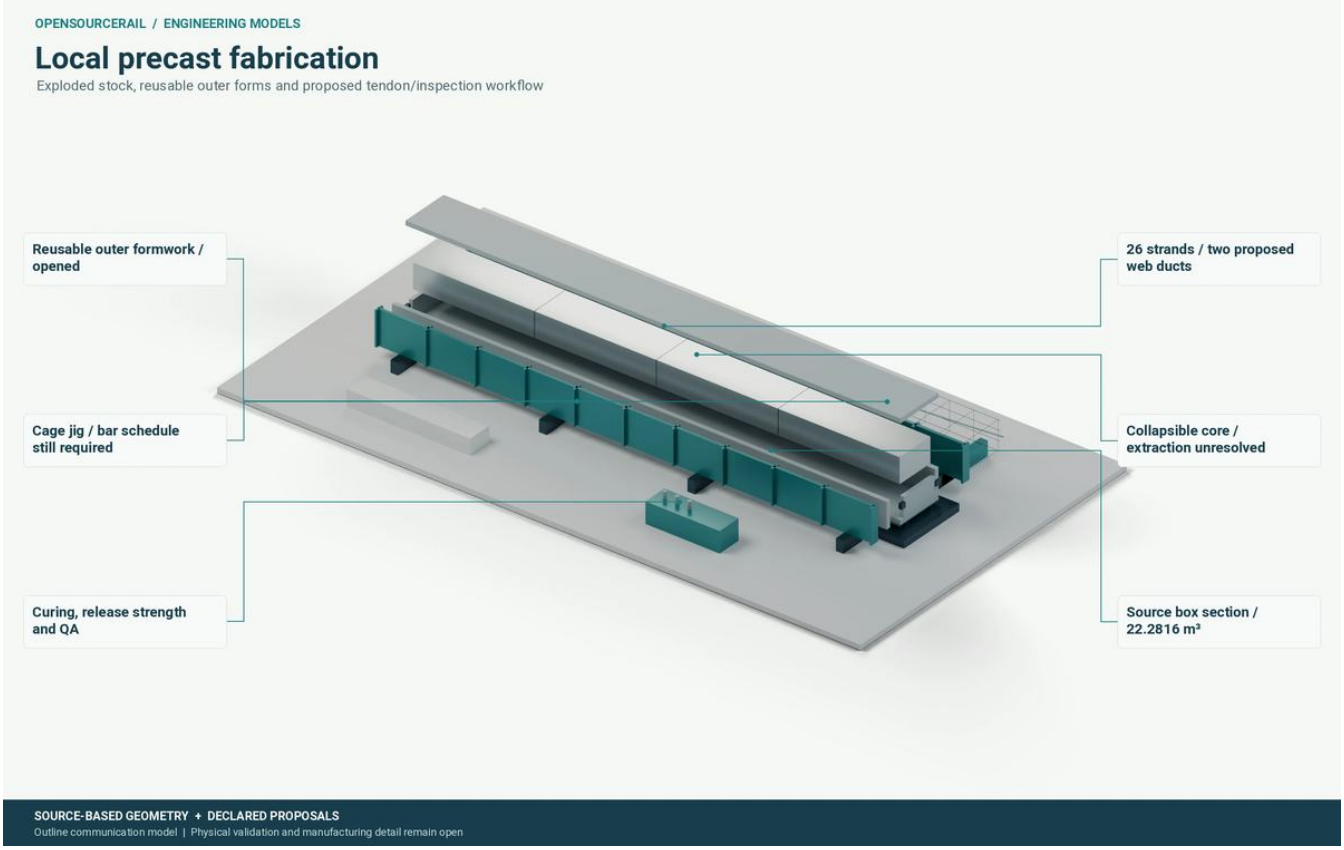
Follow loads to the ground

A train load reaches the rails, fasteners, beam or deck, bearings, piers and foundations. Each interface can govern the answer. A span that looks efficient in bending may have a difficult rail seat, bearing reaction or joint. A light beam may need expensive erection equipment or a larger maintenance closure. Moving axle positions, crowd cases, braking forces, temperature, fatigue, drainage and access must be considered together.

The project has explored a representative 20-metre precast span and several alternative civil forms. One recent numerical control used the reference train's 24 axle positions, a declared dynamic factor and finite bearing patches. Two independent mechanics routes gave very close displacements in that model; the fine result was about 10 millimetres. A refinement difference of roughly 2.7% passed a diagnostic threshold, while mesh independence, material and joint behaviour, actual rail support and site foundation evidence remained open. Agreement between solvers is valuable, but both can agree on a model that omits a real load path.

Construction is part of the design

The precast concept asks a local yard to make repeatable beams with controlled forms, reinforcement, prestressing where selected, curing, lifting points and inspection. Transport then constrains length and mass. At the site, the crane must have enough capacity at the actual lift radius, after allowing for rigging, ground bearing and the temporary state of the span. A nominal crane capacity at zero radius does not answer that question. Bearings, joints and rail installation follow, along with the inspection and replacement route.



Precast fabrication and reusable tooling concept

The illustrated yard shows reusable forms and proposed handling. It does not establish production rate, qualified prestressing or finished beam cost.

An overhead launching gantry is an alternative where a conventional crane cannot work easily. It changes temporary works, plant cost and construction sequence. Both methods need a complete erection study before one is selected.

Stations add their own structures and interfaces: platform loads, lifts, stairs, fire and evacuation routes, shelters, power equipment, maintenance rooms, drainage and public-realm connections. Their footprints cannot be inferred from a single platform length. A compact station is only affordable if access, safety and constructability survive the full design. The Samawah station pair illustrates why even a small map edit can open these larger questions.

IV. Making and keeping the railway

Local production, whole-life decisions, control and maintenance.

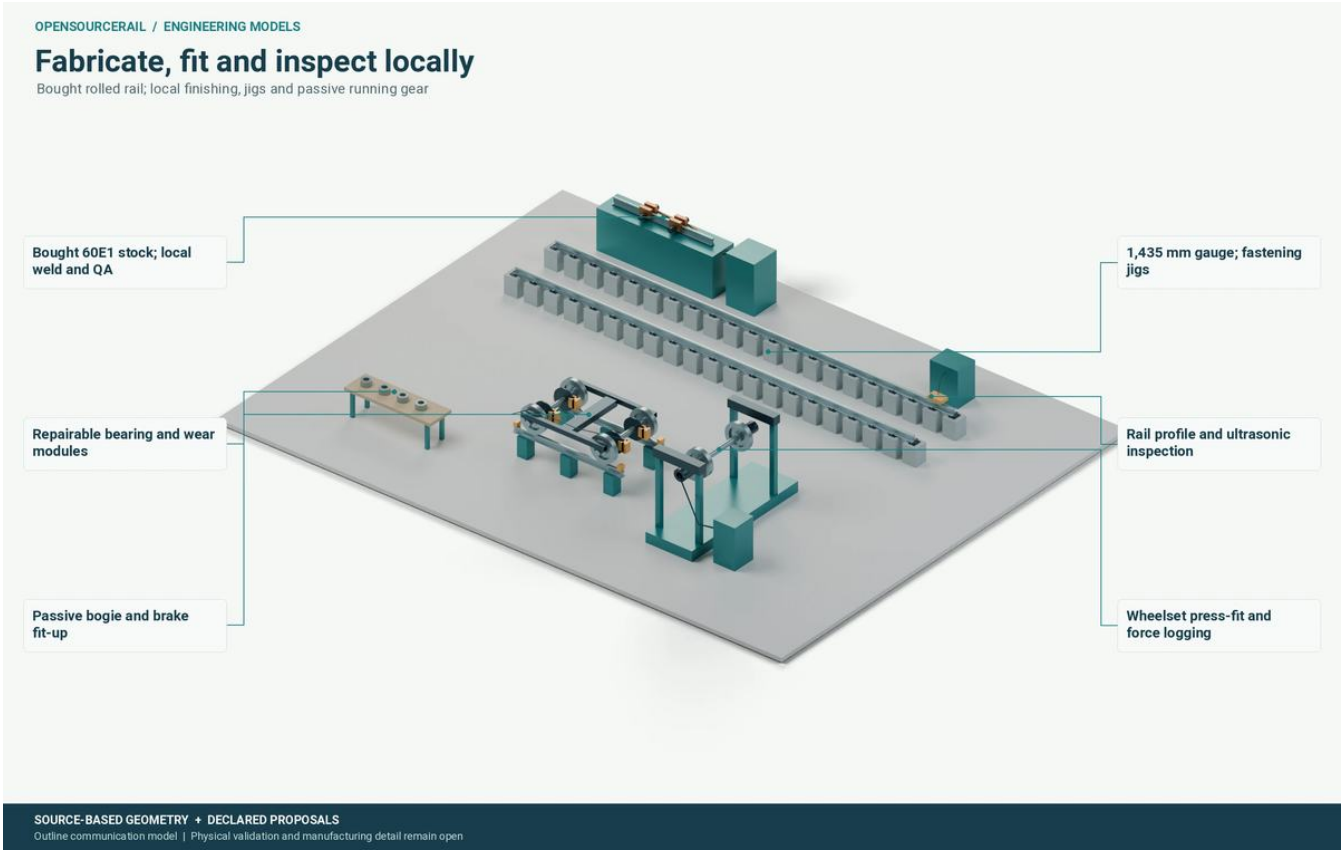
4.1 Local manufacture and sourcing

A railway can be designed for local work without pretending every specialist part will be made locally. The useful question is which knowledge and production steps can be established, inspected and sustained in a particular place. OpenSourceRail's vehicle inventory and civil studies provide a starting bill of materials. The next step is to turn each relevant item into controlled stock, process, assembly and quality information.

Make, buy and qualify

A structural member may begin as an extrusion or folded sheet. Its useful description includes grade, stock size, cut length, joining method, surface treatment, dimensional datums and inspection. A bought bearing needs a qualified specification and an incoming check, even if its internal design is left to the supplier. A door sensor or battery cell adds electrical, environmental and safety requirements. The make-or-buy choice should therefore be recorded item by item, including original material and component origin, instead of applying one local-content percentage to the train.

Candidate processes include concrete precasting, metal cutting and forming, structural fabrication, composite panels, interiors, wiring and final assembly. Proposed local capability does not itself prove a factory's yield, rate, tooling cost or skill base. A workshop may be able to make a prototype while still needing investment and training to supply a whole railway. The cost comparison should include that investment and the value of later repair and renewal work.



Proposed fabrication and inspection workshop

Illustrated workstation. Wheel, bearing and bogie manufacture or inspection would require controlled processes, qualified equipment and recorded checks.

From stock to an assembled train

The current six-car breakdown identifies 605 named leaf-part types in 40 assemblies. It includes the running gear, car structures, seats, doors, glazing, HVAC, service batteries, lighting, wiring and many joints and fasteners. That level of counting makes hidden work visible. It does not replace the next stage: a drawing that establishes mating dimensions and tolerances, an operation sequence, a jig, an inspection plan and a safe removal route.

A plausible vehicle assembly starts with a datum-controlled underframe, then adds body frame, roof and floor, bogies, equipment, panels, doors, glazing, interior and inter-car connections. At each stage the assembly must still be accessible for inspection. A late-discovered interference between a cable tray and a door mechanism is much more expensive than a revised source model before stock is cut. This is why the candidate's geometry, inventory and work instructions need to agree.

For the representative door module, the count of 694 pieces per opening is useful only if the header, carriage, guide, drive, sensor and body interfaces can be made and assembled as described. The 97 proposed locally fabricated pieces are a production opportunity, not a completion certificate. Standard purchases and qualified purchases have distinct incoming requirements.

Repair is a production route in reverse

A module's design should also describe how a technician diagnoses it, isolates power, removes it, fits a replacement, checks it and returns the train to service. The same drawings and part identities should support both factory assembly and depot repair. This helps a local operator keep knowledge after the original builder has left and gives later suppliers a clear interface to compete against.

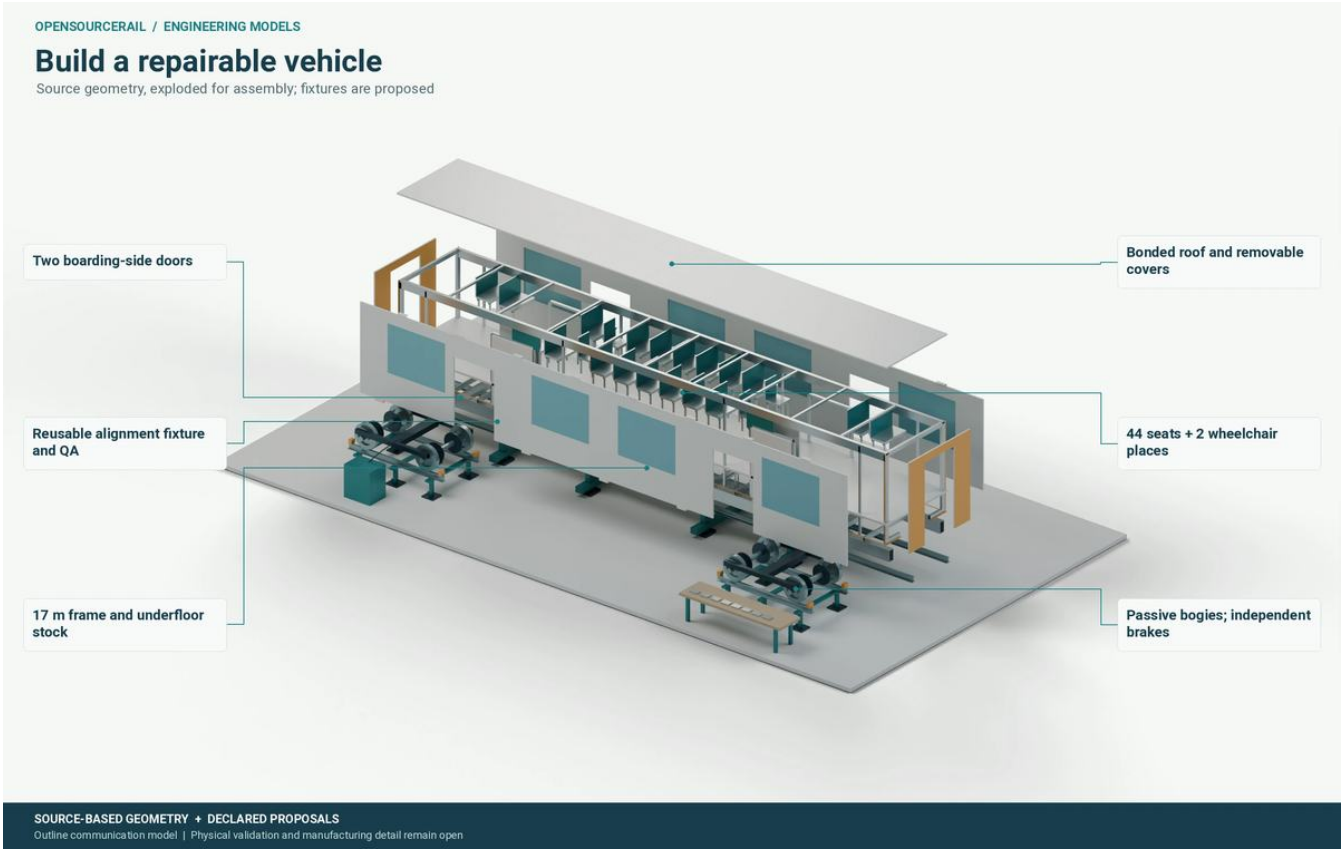
4.2 From factory to finished corridor

An affordable railway is partly a construction problem. A design that uses little material can still be expensive if it demands scarce equipment, specialized long-distance shipments, repeated traffic closures or difficult site adjustment. OpenSourceRail aims to make vehicles, stations and civil parts that local teams can produce, inspect, install and eventually repair. That objective has to be tested against the actual site and available workshops, not inferred from the apparent simplicity of a drawing.

Decide what is made locally

Local manufacture begins with a useful division of work. Some parts may be cut, welded, machined or cast in regional facilities. Others are better purchased as qualified standard items. Safety-relevant bought parts may need supplier testing, traceability and incoming inspection even when they look routine. A train bill of materials must carry these choices for every physical item, including joints, fasteners, seals, wiring, cooling and protective equipment. Omitting the small parts makes both local-content and cost estimates misleading.

The project has a detailed vehicle inventory and proposed process routes. It identifies the door operator as a particularly useful representative module: local fabrication, standard purchase, qualified purchase, electrical integration and future replacement all meet in a compact assembly. The counts are a starting map for work packages. Before manufacture, workshops need released drawings, mating dimensions, material and process specifications, inspection points and a way to record deviations. The fabrication sequence must match the actual vehicle structure, including distortion control, corrosion protection and safe lifting.

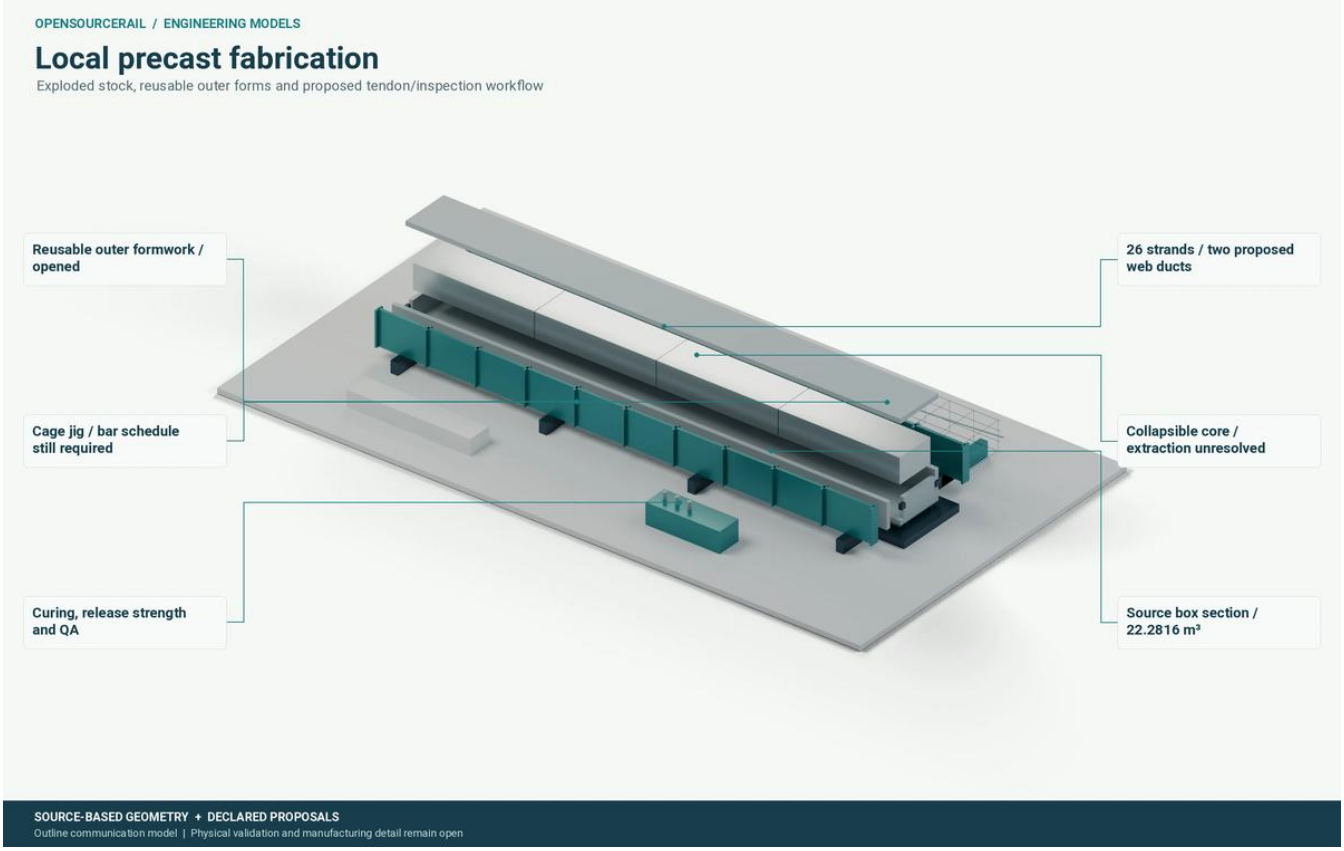


Vehicle fabrication concept

A shop can make and repair substantial portions of the railway, once the design and quality process are proved for its equipment and staff.

Precast work changes the site problem

The civil studies explore repeatable precast guideway components. Casting off site offers controlled dimensions and inspections while potentially reducing work above a busy road. But the finished element still has to be transported, lifted, supported, connected and maintained. Its production size should follow road access, factory moulds, lifting points, transport constraints and the crane that can reach the installation position.



Precast fabrication concept

A repeatable span is only one part of a complete support and erection system.

For a representative span, the construction method should state the temporary support, lifting gear, crane radius and capacity at that radius, traffic or railway closures, bearing installation, post-tensioning or other connections, survey adjustment and safe release of the lifting gear. The span also needs a load path through piers and foundations for normal, crowd, wind, accident and construction conditions. A finite-element result for the finished beam cannot establish the stability of a half-built span.

Delivery is a sequence of approvals and handoffs

The city's outline comes first, followed by survey, geotechnical work, land and utility decisions, a matched service and vehicle candidate, and construction staging. Procurement should identify long-lead items and alternative suppliers while preserving the approved technical interfaces. Factory tests, site inspection, commissioning and staff training then establish whether the built system matches the design. Passenger service requires its own independent safety and operational approvals.

The platform's project and business tools can help keep these handoffs together. A proposed bill of materials can become a production plan, a received component can be inspected, and a fitted module can enter a maintenance record. Existing demonstrations show parts of that digital route, including a synthetic business-system example. The full candidate linked transaction for a real door module remains to be completed. The software should make omissions and conflicting revisions visible before someone orders, builds or fits the wrong part.

Repairability affects the first drawing

A replaceable unit needs reachable fixings, a safe isolation procedure, clearance to remove it, lifting provisions if it is heavy, and enough space to inspect its neighbours. These are geometric facts, not afterthoughts for a manual. A door guide that cannot be extracted through the available opening may force a larger body change. A power cabinet that blocks a crane route may change a station layout. Testing repair sequences early can reduce future service disruption and can inform the cost comparison before a design is selected.

4.3 Cost and whole-life decisions

Affordability is a property of the complete service, not a single vehicle or span price. A city needs to know what it would spend to design, build, equip, operate, maintain, renew and eventually remove the railway. It also needs to know what passenger capacity and journey quality those costs buy. A fair comparison holds the service objective and its safety obligations constant while allowing competing designs to meet them differently.

Build a price from physical work

A useful estimate follows quantities and production routes. A beam needs concrete, reinforcement or tendons, forms, labour, curing space, lifting, transport, bearings, erection and rail fixing. A carbody panel needs stock, cutting, forming, joining, finishing, fasteners, inspection and replacement planning. Bought specialist equipment needs a specification, delivery, integration, testing and eventual renewal. A unit price with no scope definition can hide any of these steps.

The platform keeps factory, delivered, installed and whole-life costs separate. It compares low, base and high design-stage scenarios with explicit rates and allowances. Fourteen cost scopes connect different parts of the railway, including rolling stock, guideway, stations, propulsion and power, construction, operations, maintenance and finance. A rate borrowed from another city is a provisional assumption. A supplier quotation can strengthen one account without making all the other accounts certain.

Cost boundary	Examples of questions
Factory	How much stock, labour, machine time, tooling and QA does the part need?
Delivered	What are packaging, transport, duties, handling and supplier terms?
Installed	What civil works, wiring, lifts, testing and site access are needed?
Whole life	How often are parts renewed, what energy and staff are needed, and what remains at disposal?

The vehicle's detailed parts proposal has a base direct parts-and-process subtotal of about $1.064 million across six cars. It excludes final assembly, shared plant, delivery, qualification, warranty and many whole-life accounts. It is not the purchase price of a finished train. Likewise, the Samawah comparison's base installed capital is about $1.14 billion and its whole-life present cost about $2.26 billion under its own planning assumptions. Neither figure is a construction bid or a city budget.

Price service, not just hardware

Fleet size follows route time, frequency, availability and spare policy. A station change can alter energy or access even if it leaves the number of operational stops and trains unchanged. A different car formation can alter platform length, depot space and civil loads. A powered-wheel train might reduce station equipment but increase vehicle mass and maintenance. These cross-effects are why each economic result must retain its physical and service candidate.

Current scenarios use declared production and civil shift patterns and separate operating-hour cases. They are planning assumptions, not an approved roster or construction schedule. Wages, productivity, supplier prices and ground conditions can all change a ranking. The model should show where a change matters most, so the next survey or quotation is chosen for its decision value.

Affordability is not achieved by erasing a mandatory safety item, calling an unknown mass zero or postponing a required test outside the cost boundary. A more honest early estimate may be wider, but it gives the owner a sounder basis for reducing cost through design, production and service choices.

4.4 A corridor cost comparison

Costs become more useful when tied to a service a city could recognize. OpenSourceRail retains a thirty-kilometre elevated, double-track comparison with twelve stations. It assumes a 72 km/h speed limit, one metre per second squared acceleration, a two-minute peak interval and 276 named places per reference train. The calculation uses 52 six-car trains, including its fleet and spare assumptions. This is a design-stage benchmark, not a priced order or a decision to build that particular corridor.

Two prices can describe different assumptions

One generic-rate Class C central estimate is about US$738 million, or US$24.6 million per route-kilometre. It includes trainsets, stations, civil works, a provisional compact launch and power concept, depot, production and tooling, testing and contingency under its declared rates. A separate project-volume supply comparison reprices the same corridor and fleet using wholesale and manufacturer observations and calculated manufacturing alternatives. Its central installed estimate is about US$548.1 million, or US$18.27 million per route-kilometre. Shared premises, freight, quality work, duties and future replacement setup are counted in that comparison.

The difference is not a verified saving available to every city. The two studies test price and supply assumptions, while the physical design and service control remain provisional. Neither has supplier quotations for a complete qualified railway or a surveyed site. A local owner would need to replace transferred prices, ground assumptions, access, land and utilities, construction method, tax and financing terms with its own evidence.

What is held in the benchmark	What a new city must establish
Thirty-kilometre, twelve-station study scope	Actual route, access, land and ground
Reference train, places and two-minute control	Demand, operating plan and protected headway
Provisional drive and civil families	Candidate-specific equipment and structure
Scenario rates and model quantities	Supplier bids, local labour and site quantities

Use a benchmark to ask better questions

If civil work dominates the price, a city may investigate span form, station placement, construction access and utility diversions. If rolling stock or propulsion dominates, it can compare formations and drive families at equal passenger service. If the fleet is large because round-trip time is long, a station or route change might have more value than a small saving on each car. The point of a model is to direct evidence-gathering and design effort toward the choices that matter most.

The Samawah case illustrates a different scale of question. Its midpoint station edit barely changes the modelled capital total yet increases the whole-life present cost through energy. That is not contradictory. Capital, operating cost and passenger usefulness respond to different parts of the design. A lower first cost does not guarantee a better whole-life choice.

Affordability should be measured alongside journey quality, availability, accessibility and local support. Mandatory safety and passenger protection are constraints on every candidate. The benchmark's figures make the discussion concrete; they become investment evidence only when the selected physical candidate, service, site, suppliers and risks have been reviewed together.

4.5 Control and passenger protection

A safe passenger railway needs more than a train that can move under normal conditions. It needs clear authority for every movement, protection from conflicting movements, reliable detection, a response to faults and means to bring people to safety. OpenSourceRail researches open control software alongside the physical railway, but the current software is a research and simulation platform without live movement authority.

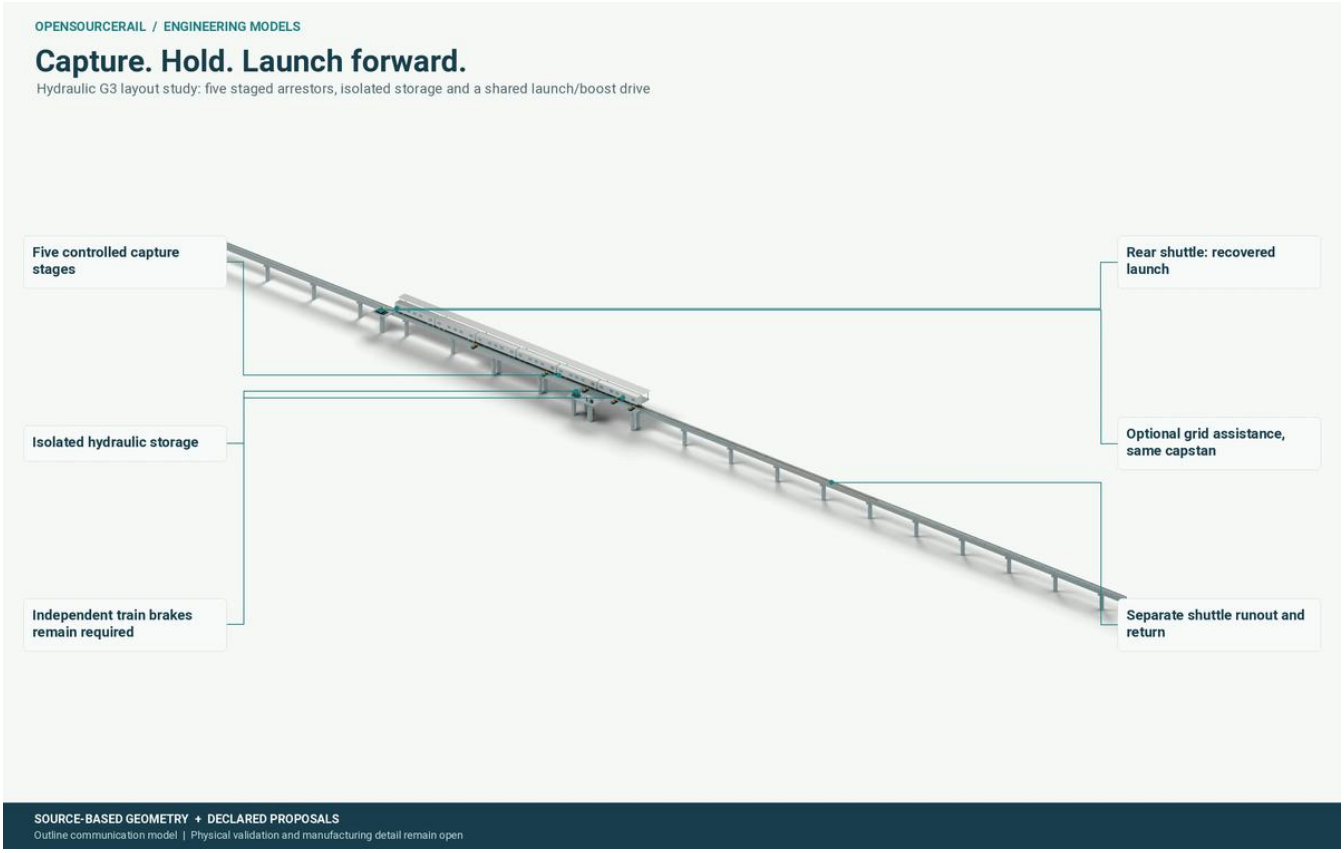
Separate normal service from protection

The service plan describes when a train is intended to depart, stop and arrive. Protection limits where it is permitted to move and how it must stop if the service plan or equipment fails. Station launch equipment may drive a train in the normal journey, but independent onboard brakes must remain able to stop it when station power or communication is unavailable. A protected headway requires validated braking, position, supervision, platform and terminal assumptions, not just dividing an hour by a planned interval.

The reference system must consider train detection, route setting, station dispatch, doors, platform interface, overspeed, emergency stops and recovery. Degraded cases include a missed station handoff, a stopped train, sensor disagreement, loss of power or an occupied section. The response has to fit the actual physical train and route. A correct software test with idealised braking data does not prove the corresponding wet-rail stop.

Open control needs a clear boundary

Rust research software provides a place to define state machines, interfaces and deterministic checks. Simulations can expose contradictory authority, unsafe transitions and unexpected restart behaviour. They can be reviewed independently and rerun as designs change. The resulting evidence is valuable for engineering, but deployment would need qualified sensors, actuators, communication, operating procedures, safety analysis, testing and the approvals applicable to the actual railway.



Station equipment and launch concept

Communication model of a station-side equipment arrangement. The final drive, protection hardware and maintenance access are candidate-specific.

FUXA and the project gateway are explored for equipment observations, alarms, trends and review-linked records. Observation is not the same as commanding a train. A live deployment would need authenticated device bindings, a commissioned network, permissions, fail-safe interfaces and maintenance procedures. The Workbench is a design and review interface; it cannot grant movement authority.

Passenger protection must also shape the physical station and vehicle. A control response is only useful when the doors, platform, evacuation route, communications and rescue method allow people to be cared for. The project therefore keeps control, operations, braking and civil design attached to the same candidate and the same emergency scenarios.

4.6 Business, maintenance and renewal

A railway lasts through many generations of staff and equipment. A component must be received, inspected, installed, maintained, replaced and eventually retired. The information that follows it can make those tasks safer and less expensive. OpenSourceRail connects engineering records with proposed manufacturing, procurement, quality and maintenance workflows so that a physical revision is visible beyond the design office.

One component, one history

Consider a door operator. Its drawing defines its fit; its bill of materials names the parts; a purchasing record identifies the supplier; incoming QA checks the received item; assembly records where it was installed; maintenance records faults, repair and replacement. A technician should be able to see which revision is on the train and which inspection limits apply. If a supplier changes the motor or a body mount moves, affected work instructions and spare parts must change with it.

The current software can prepare candidate-linked drafts and has exercised a bounded synthetic custody example. It can reject a purchased child without a serial, batch, incoming inspection or valid parent consumption record. The complete counted door module has not yet passed a real native ERPNext receipt-to-assembly-to-replacement journey. That is an integration milestone, not a detail to infer from the smaller example.

ERPNext and Frappe HR are relevant to purchasing, stores, work orders, quality, people and training. The project gateway and FUXA can support equipment observations and alarm history. These tools require a real owner, permissions, warehouse and asset masters, authenticated devices and commissioned procedures. A software demonstration does not create legal purchasing authority, accepted equipment or a railway operating organization.

Design for replacement

Maintenance planning begins before a part fails. The door needs isolation points, access, lifting or handling provisions, a removal path, spare stock and a post-replacement check. A service battery needs thermal management, fire isolation and a cassette or other removal route. A guideway bearing needs inspection and an eventual jacking and replacement method. These choices alter mass, civil space, workshop tools and lifecycle cost.

The same principle applies to renewal of software and data. A city owner should retain readable design information, open interfaces, licenses and the ability to move to another supplier. Historical calculation records should remain retrievable so a later engineer can understand why a design was selected or rejected. Current work should be kept compact enough that the evidence remains usable rather than overwhelming the ordinary checkout.

People carry the system

Production and maintenance capability rests on trained people as much as machines. Inspectors need approved criteria; technicians need safe work instructions; planners need to know the assumptions behind costs; operators need fault and rescue procedures. The project can help organize those records, but a real railway would need an accountable employer, accepted competence, independent checks and an operational safety system. The aim of the open platform is to make that work visible and transferable, not to replace it with a database entry.

V. Evidence and the road ahead

What the tools establish, what remains open and how v0.7 advances.

5.1 Engineering tools and evidence

The open engineering platform is meant to keep a railway decision coherent as it passes from city planning to geometry, structural analysis, inventory, cost, manufacture and maintenance. Its value lies in the connections. If a station moves, the dependent service and cost results should update; an unchanged vehicle model should remain available. If a carbody changes, mass, clearance, braking, civil load and factory quantities must be reconsidered.

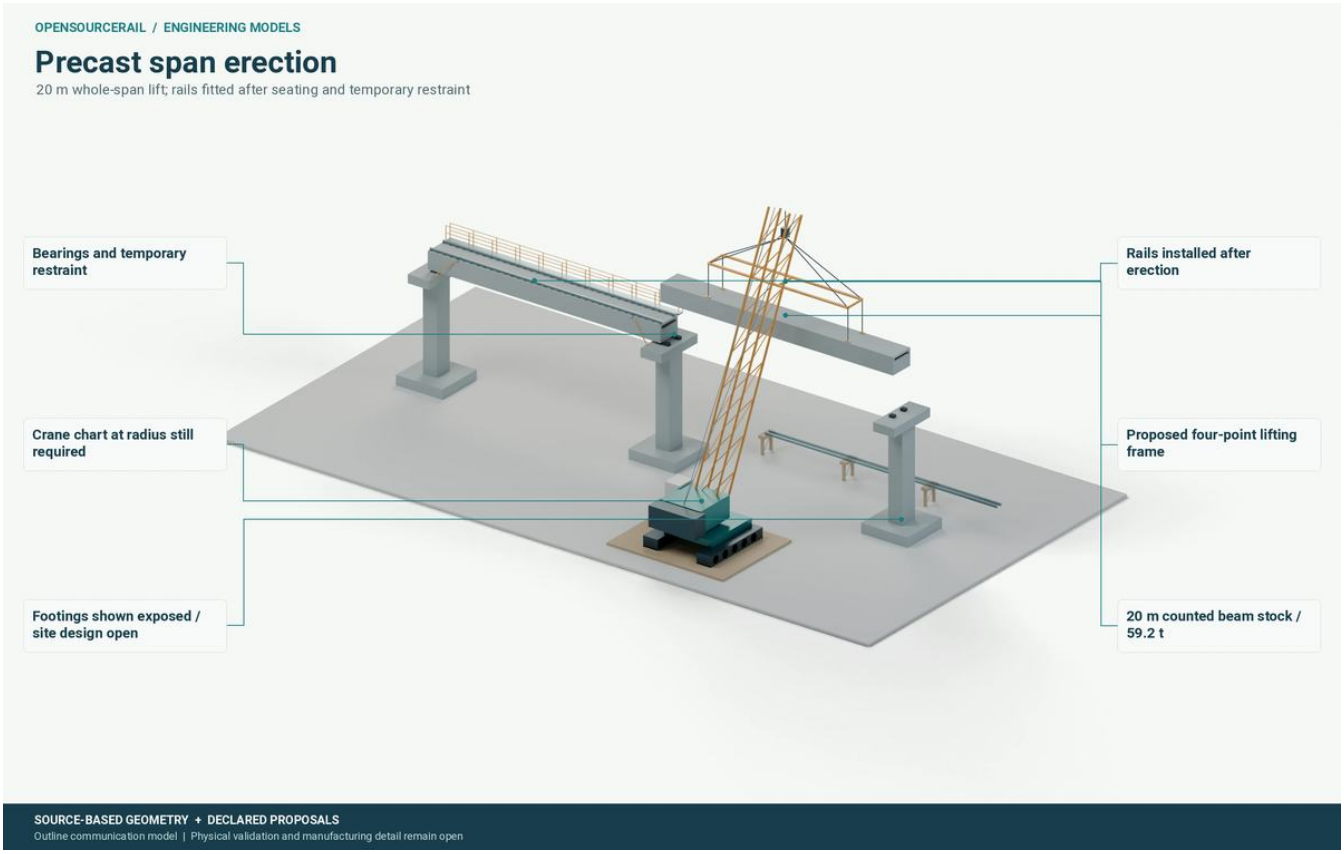
A design has more than one kind of identity

The physical candidate describes the geometry, material, components and interfaces of the proposed railway. A numerical evaluation describes the model, assumptions, loads, tool and conditions used to test it. A selection records which candidate is preferred for a stated purpose and why. Keeping these separate prevents a new calculation from pretending to be a new train, and prevents an old result from being attached to a changed design.

The tools keep both successful and failed evaluations. A body that does not meet an applicable screen can still teach the team about a load path. A nonconverged numerical solve may reveal a modelling problem. An interrupted run should not disappear, because its partial evidence and resource use can guide the next experiment. Reuse is valuable only when the relevant source, candidate, model, load and tool conditions still match.

Models at several levels

Early design uses geometry and analytic checks to discard clear definition failures cheaply. More promising candidates can progress to coarse numerical models, then refined and coupled studies. SolidCraft is being used for candidate geometry and assembly work; FieldCraft for matched mechanics studies. Independent FreeCAD and CalculiX routes provide useful comparisons where their methods apply. RustGIS carries native city edits in the tested domain. A tool's name does not establish that it can represent every joint, material law, electromagnetic transient or railway operating condition.



Engineering model of the proposed construction and vehicle system

A model helps make interfaces visible. The real lift still requires the chosen beam, crane chart, ground, rigging and temporary-works design.

CAD is particularly useful when a part must fit another part or be removed from an installed assembly. FEA can test a declared load path and compare mesh and solver sensitivity. GIS can bring a station edit back into the costed corridor. None of these operations automatically proves manufacture, structural safety or route approval. Each output must state what was represented and what remained outside its model.

A readable result

The Workbench should make the relationship visible to a planner, engineer or reviewer. One project view should show the corridor edit, chosen vehicle, civil option, quantities, costs, drawings and unresolved findings together. The reader should be able to tell whether a result is a proposal, a checked calculation, a verified software workflow or an accepted physical result. A held or stale result is still useful history, but it must be clearly marked and kept out of the supported decision path until renewed.

5.2 Current results and open questions

The v0.6.1 Open Engineering Platform is a published software release. It offers planning and research tools, a Workbench, candidate-linked examples and documented limitations. The Samawah software journey has been installed and demonstrated from the published release. That supports the claim that the software route can be used and reviewed. It does not make a railway physically accepted or a city design ready for construction.

Development since that release has joined the Samawah station edit to service, fleet, fourteen cost scopes and a revised dossier view. Native GIS edits have returned through Samawah and Baghdad calculations. Candidate studies have counted vehicle parts and door components, explored new CAD and mechanics backends, and compared independent numerical results. These are meaningful steps toward a complete engineering workflow. Their findings still belong to the exact candidates and model domains used to produce them.

Area	What exists	What still decides the physical design
Corridor	Comparative route, station, service and cost calculations	Surveyed access, demand, land, ground, utilities and chosen service
Vehicle	Counted components, candidate geometry and analytical controls	Installed fit, complete mass, joints, dynamics, fire and passenger tests
Propulsion and braking	Conditional force, energy and stopping studies	Available installed force, repeated thermal duty and validated braking
Civil works	Span and moving-load models with independent comparisons	Bearings, rail seats, foundations, fatigue and erection at the actual site
Manufacture	Proposed routes, stock, BOM and assembly order	Controlled drawings, qualified processes, supplier evidence and QA limits
Lifecycle	Draft business and maintenance handoffs	Actual candidate-linked transactions, organizations and reviewed records

A numerical result has a boundary

Two solvers can agree closely on a model while its real bearing, fastening or material behaviour remains uncertain. A finite-element mesh can refine toward a stable displacement without proving that the chosen load case is complete. A motor method can reproduce an ideal field while leaving the manufactured primary, current closure and heat transfer absent. The correct response is to keep the useful numerical evidence and name the next measurement or model needed to make a physical decision.

The current installed vehicle still needs resolved door and equipment interfaces, removal paths, mass and inertia. Propulsion needs force-speed and repeated-duty maps linked to the chosen train and corridor. Braking needs independent stopping envelopes. Civil work needs a represented rail/support interface, foundations, moving and accidental loads, and a complete erection method. Cost needs equal-service comparisons, site quantities and stronger supplier evidence. These questions are connected: a heavier fitted vehicle changes the beam and braking problem, while a changed station changes service and energy.

The older integrated reference dossier is retained with a source-binding hold. Its attractive global cards are not treated as current supported results. Historical maglev and other research remain available to study; they do not replace the wheel-supported reference by proximity in a folder or a book. This separation protects both the history and the present design.

5.3 Research choices and independent checks

The wheel-supported railway is a development reference, not the end of exploration. The project keeps earlier maglev work as historical research and allows new candidates to test propulsion distribution, train formation, running gear, speed, station spacing and civil form. The purpose is to find a useful, affordable system under real constraints. An attractive isolated result should lead to a fuller comparison, not silently replace the reference design.

Compare the service that a city receives

A lighter carbody, shorter train or different drive may save capital while changing seats, frequency, power, platform length, repair effort and failure response. The comparison is fair only when each candidate is asked to deliver the same defined transport service and mandatory passenger protection. If a concept offers less capacity, takes longer or needs more vehicles, those differences belong in the result. Local manufacture and repair capability are also part of the objective, because a system that cannot be supported locally may become expensive or unreliable later.

Different questions need different models. A quick analytical calculation can reject an impossible geometry or expose a missing mass account before a large numerical study. A coarse structural model can identify important load paths. Fine models are useful when their mesh, material, supports and loads represent the decision at hand. Physical tests are needed when a model cannot establish a critical interface or material behaviour. Moving up this ladder costs time and storage, so the most valuable question is often what evidence would change the choice between two candidates.

Agreement is useful, with a boundary

Two independent solvers that agree on a span's displacement give confidence that the numerical implementation of that particular model is consistent. They do not prove that the real rail seat, bearing, pier and foundation have the same behaviour. Repeating a mesh at greater resolution can reveal numerical sensitivity. It does not introduce a missing bolt, fatigue case or construction load. A physical measurement, with an approved test method and independently reviewed limit, has a different role again.

The civil work already contains comparisons between newer SolidCraft and FieldCraft methods and independent FreeCAD and CalculiX checks. The new tools can help agents generate geometry and evaluate alternatives, while the older independent route remains valuable where it tests the same physical question through another implementation. A tool should be retired only when its role is either replaced with evidence or clearly preserved for historical reproduction. Convenience alone is a weak reason to lose an independent check.

Keep failures useful

A failed design can teach the project where a load path, interface or model is inadequate. A solver that does not converge, a candidate that exceeds a thermal limit and a mechanism that cannot carry the declared axial load are different findings. Each should retain its inputs, conditions and reason for failure. A rejected pin-jointed mechanism does not reject a bonded shell unless the shell is tested under an appropriate model. An unsupported material law should remain unsupported rather than be replaced quietly by a convenient substitute.

Research alternatives also need ancestry. A change in magnetic-flux hypothesis may be a new evaluation of the same physical geometry. A new bogie or door layout is a physical revision and should propagate through mass, braking, civil loads, manufacturing and cost. A selection records why one candidate is preferred for a stated purpose; it is not a claim that the railway has passed physical acceptance. This discipline lets an open community explore boldly while keeping each conclusion understandable.

5.4 One complete candidate for v0.7

The immediate v0.7 target is a journey through one project. A user changes a Samawah station or alignment and sees the resulting service, fleet, civil, energy, cost and dossier changes. From that same project, they can open one vehicle module and follow its geometry, manufacture, inspection and maintenance. Every result should refer to the same candidate or explain why it is a separately evaluated alternative.

The corridor and its train

The Samawah midpoint comparison is the first working step. The next work should bind the corridor to one selected six-car installed candidate, one representative civil and station arrangement, and a propulsion/braking evaluation that can support the proposed journey. A station edit should recalculate only its dependants. A vehicle change should trigger the mass, force, structure, manufacturing and cost work it actually affects. The Workbench should show those changes and the open site inputs without asking a reader to inspect a directory of numerical archives.

Baghdad provides the second city test. Repeating the same method on a different corridor checks that the project is not tuned only to one set of Samawah assumptions. It also exposes which rates, route conventions and station interfaces are genuinely shared and which must be replaced with local information.

One module through its life

The door operator is the representative module. Its inventory already reveals much of the required work; v0.7 needs a mated, installed design with controlled header, rail, carriage, drive and sensor dimensions. Door motion and removal must fit the vehicle. The same module should then have a manufacturing drawing set, tolerances, tools, assembly operations, incoming inspection and service replacement instructions. Its actual candidate-linked bill of materials can be carried through business and maintenance records.

This is a deliberately narrow demonstration. A full train has many other critical systems, and physical passenger safety still requires more than a working software journey. One complete, reviewable path through the tools is more valuable than thousands of superficially filled records.

Engineering work that follows

The vehicle team must close installed mass, equipment access, joints and running gear. The propulsion team must establish available force, electrical and thermal duty, and independent stopping. The civil team must model the actual rail and support interface, bearings, foundation and erection. Cost and service work must compare alternatives at equal passenger service with the same candidate quantities. Physical tests, supplier information and independent review have their own milestones.

The platform's tool choices should follow that work. A measured slow stage may justify more Rust or vectorisation, while a dependency error should be fixed before making the wrong calculation faster. SolidCraft, FieldCraft and RustGIS can become preferred where they have demonstrated the required geometry, mechanics or editing domain. Other tools remain valuable for independent checks until an equivalent route has been shown. Historical research stays retrievable even when an obsolete default installation is retired.

The outcome sought for v0.7 is a comprehensible comparison and a genuine production handoff, with its unresolved evidence plainly stated. That is the right foundation for a physical prototype and, later, a railway that a city can understand, build and maintain.

5.5 Sources, terms and licensing

This is a reader's edition of the current platform and engineering work. The separately published source receipt records the exact files and illustrations used to build the PDF. The maintained project guides provide the detailed calculations, software behavior and historical records behind the narrative. The following titles are useful starting points for further reading; they remain subject to their own stated evidence limits.

Further reading	Main subject
OpenSourceRail platform guide	Three products, passenger railway concept, delivery method and local capability
v0.6.1 release and known limitations	Published software scope, installation and visible holds
v0.7 Samawah corridor continuity	Before-and-after station comparison and its planning boundaries
v0.7 execution and evidence	Current development results and unresolved interfaces
Vehicle parts, manufacture and assembly	Counted stock, proposed processes and assembly breakdown
One coordinated railway experiment	Same-candidate engineering propagation and research alternatives
SolidCraft and FieldCraft backend pilot	CAD and mechanics comparisons and their model domains
v0.7 corridor-to-lifecycle roadmap	The integrated work still needed

Terms used in this book

Candidate means one physical arrangement of the railway and its relevant interfaces. Evaluation means a calculation or test under declared conditions. Selection means a decision to use one candidate for a stated purpose. Source-bound means that a result identifies the inputs and method to which it applies. Whole-life cost includes construction, operation, maintenance, renewal and end-of-life work under an explicit period and discounting assumption. Installed mass includes the equipment, mounts, wiring, cooling, protection and other items actually needed on the vehicle, beyond a convenient stock subtotal. Physical acceptance would require appropriate measurements, approved limits and independent review; it is not conferred by a software demonstration.

Images marked concept communicate intended form or use. Engineering model views illustrate declared geometry and methods, but are only evidence for their exact model and checks. No image in this book is an approved manufacturing drawing, surveyed railway route or operating permission.

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